

BFS Traversal

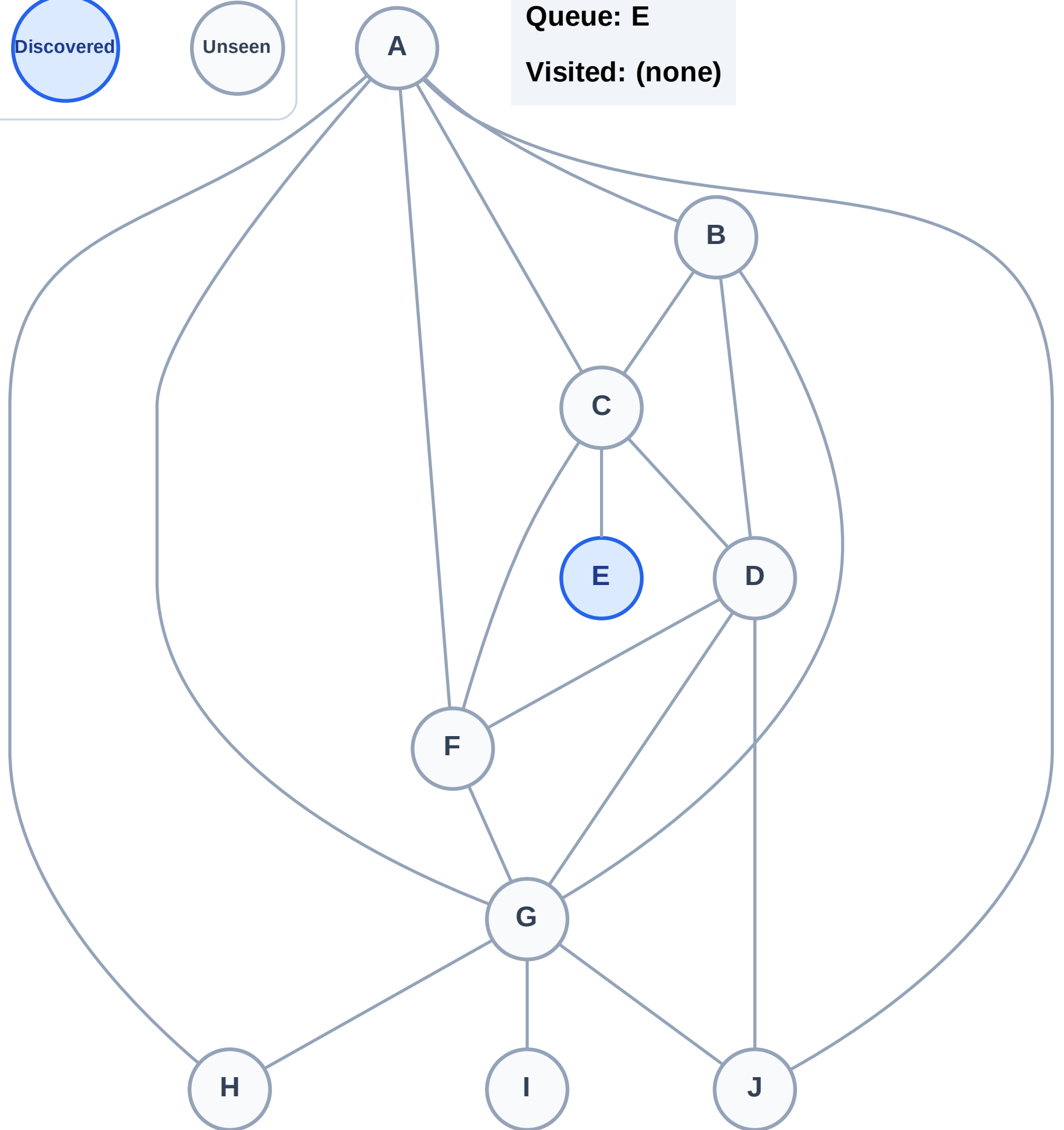
Step 1: Start at E

Legend



Queue: E

Visited: (none)



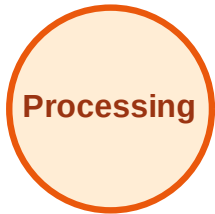
BFS Traversal

Step 2: Process node E

Legend



Complete



Processing



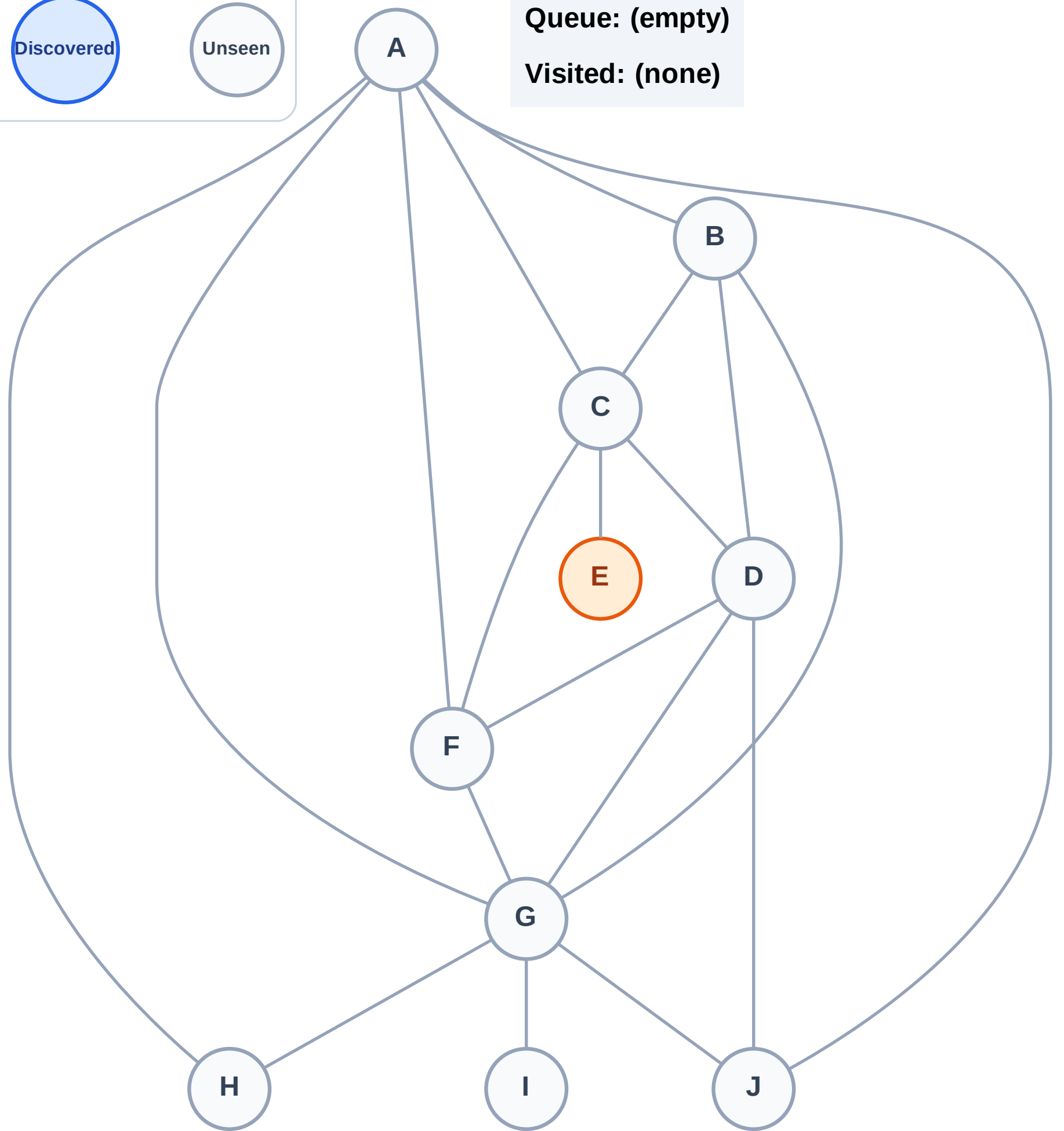
Discovered



Unseen

Queue: (empty)

Visited: (none)



BFS Traversal

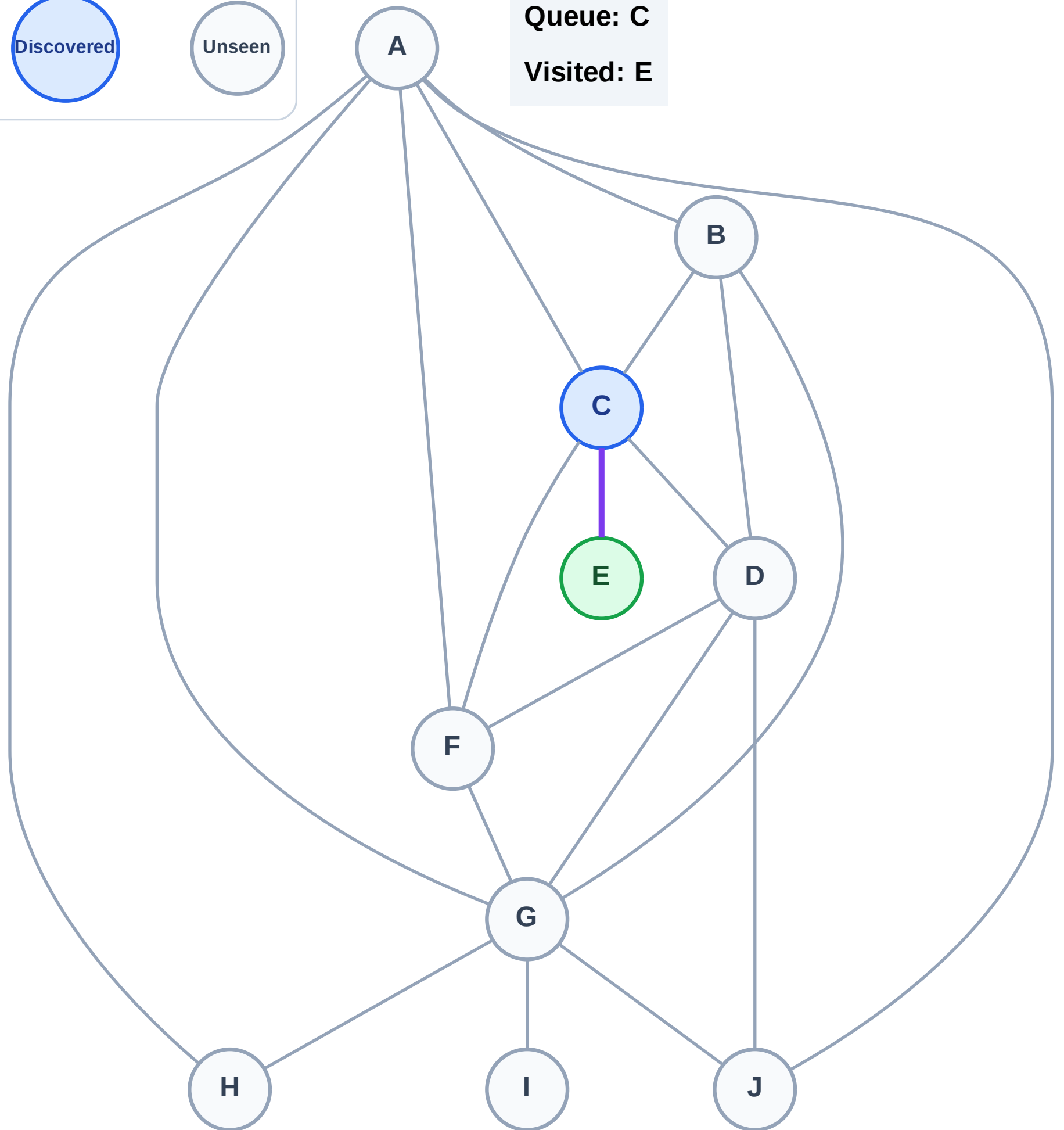
Step 3: Discover C from E

Legend



Queue: C

Visited: E



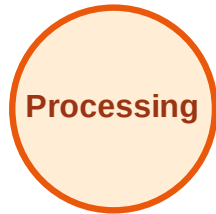
BFS Traversal

Step 4: Process node C

Legend



Complete



Processing



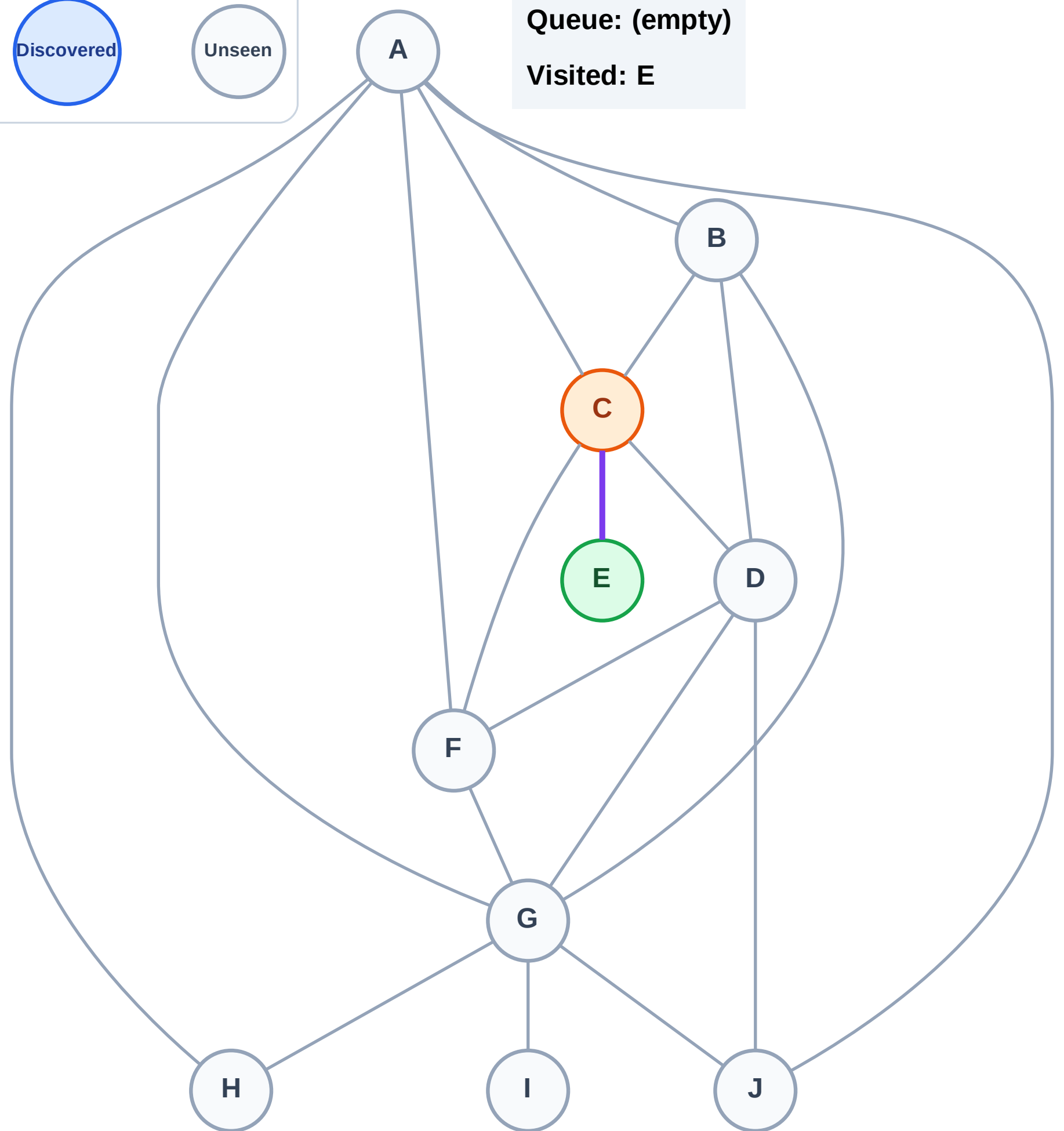
Discovered



Unseen

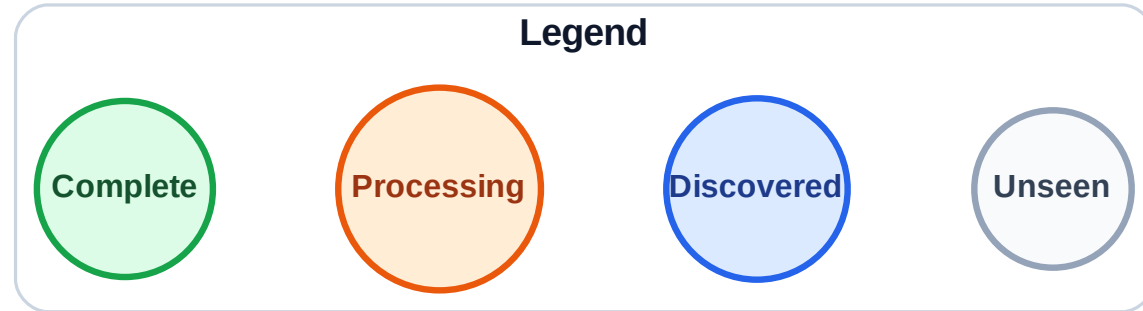
Queue: (empty)

Visited: E



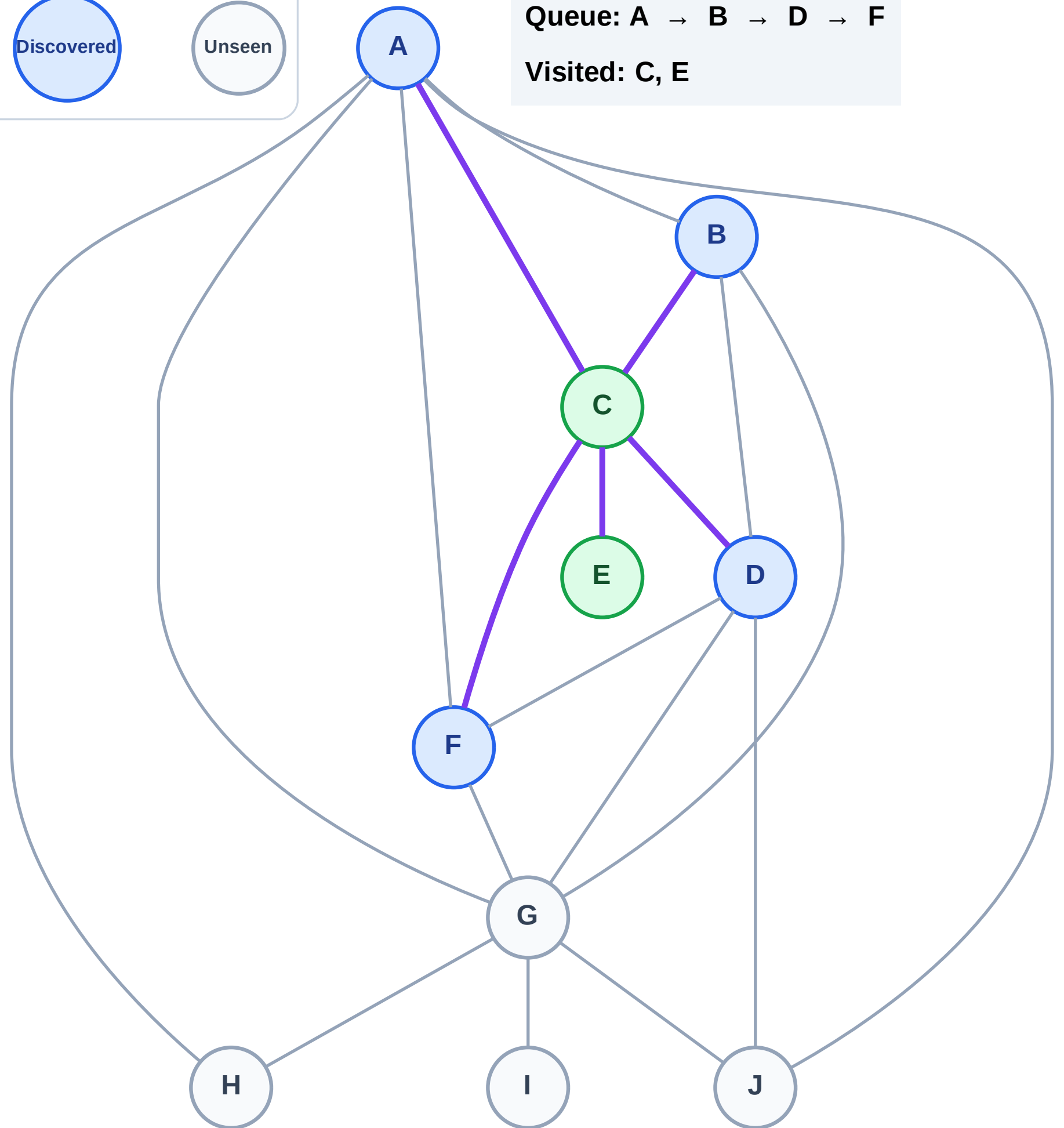
BFS Traversal

Step 5: Discover A, B, D, F from C



Queue: A → B → D → F

Visited: C, E



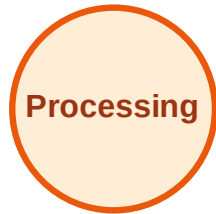
BFS Traversal

Step 6: Process node A

Legend



Complete



Processing



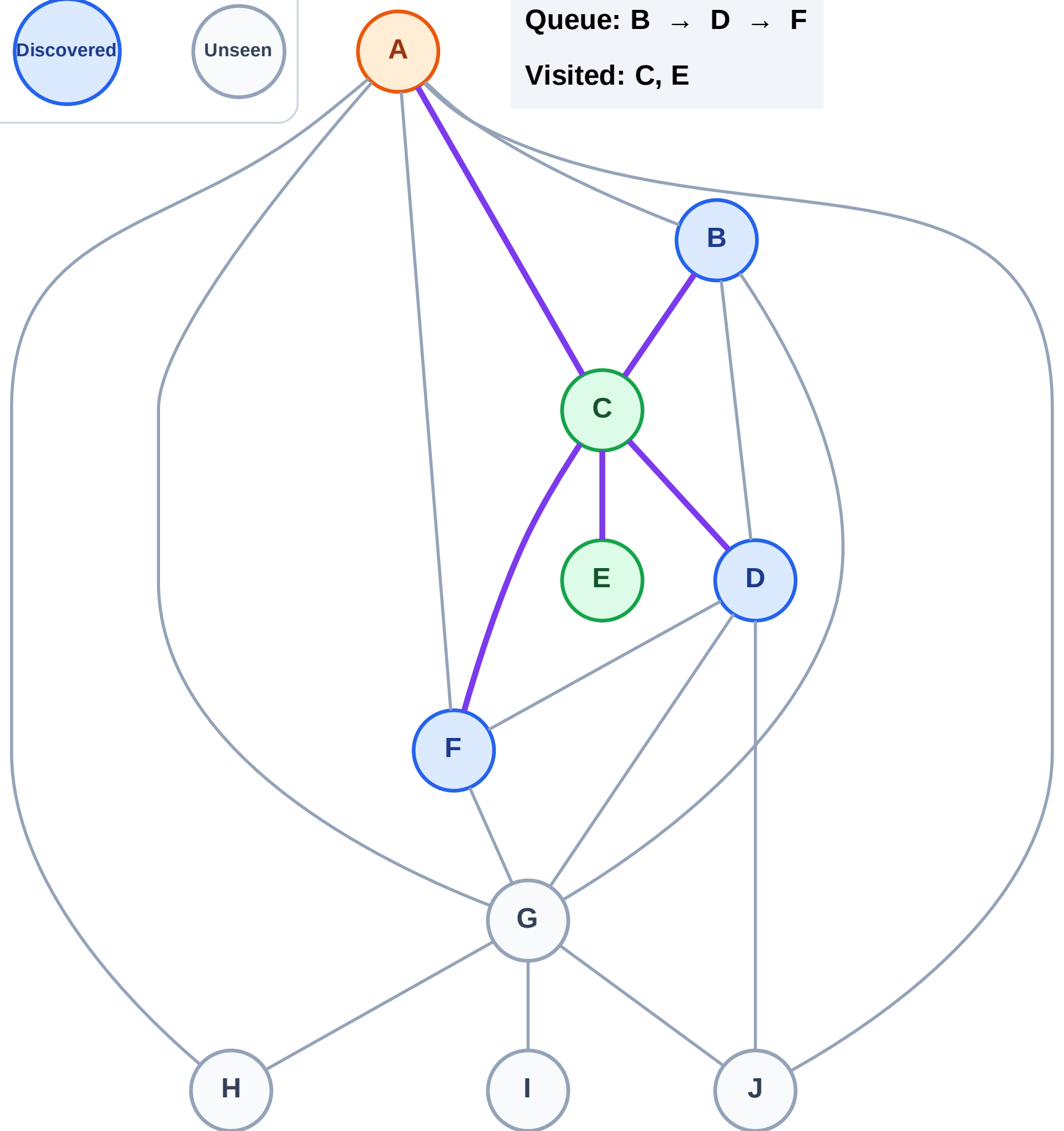
Discovered



Unseen

Queue: B → D → F

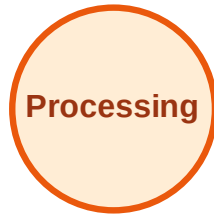
Visited: C, E



BFS Traversal

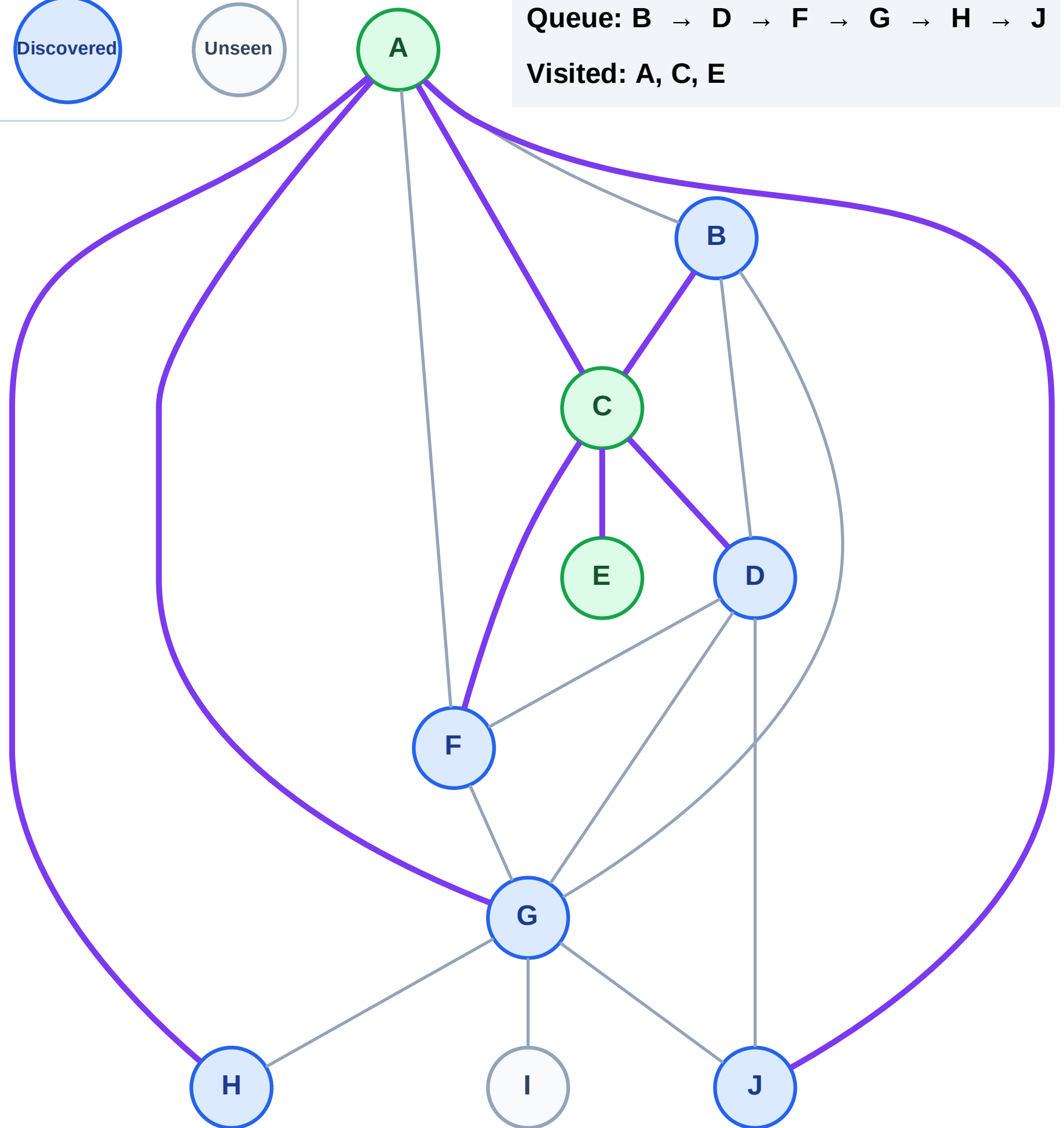
Step 7: Discover G, H, J from A

Legend



Queue: B → D → F → G → H → J

Visited: A, C, E



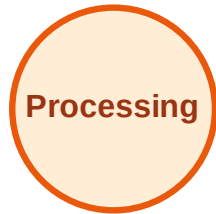
BFS Traversal

Step 8: Process node B

Legend



Complete



Processing



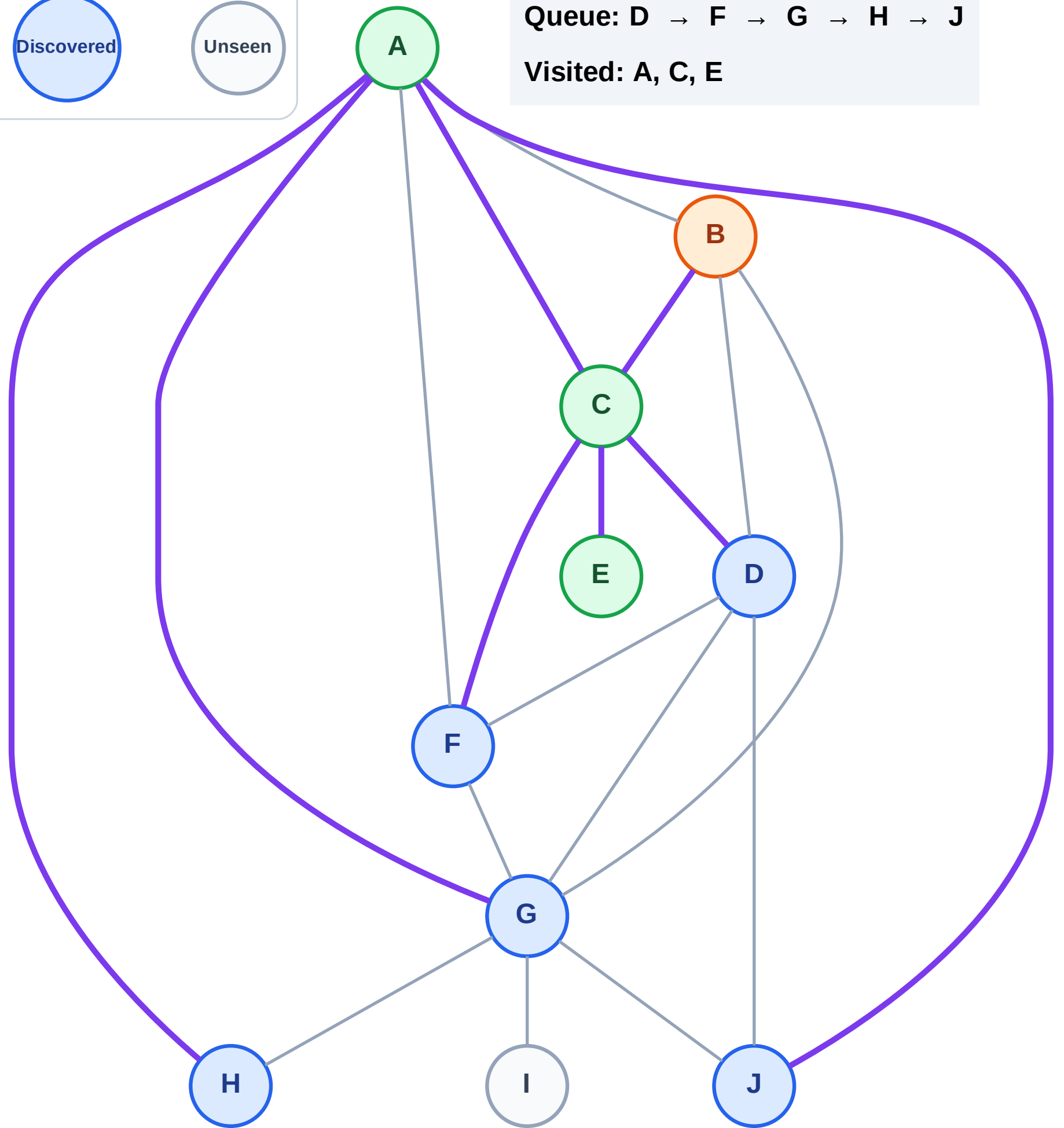
Discovered



Unseen

Queue: D → F → G → H → J

Visited: A, C, E



BFS Traversal

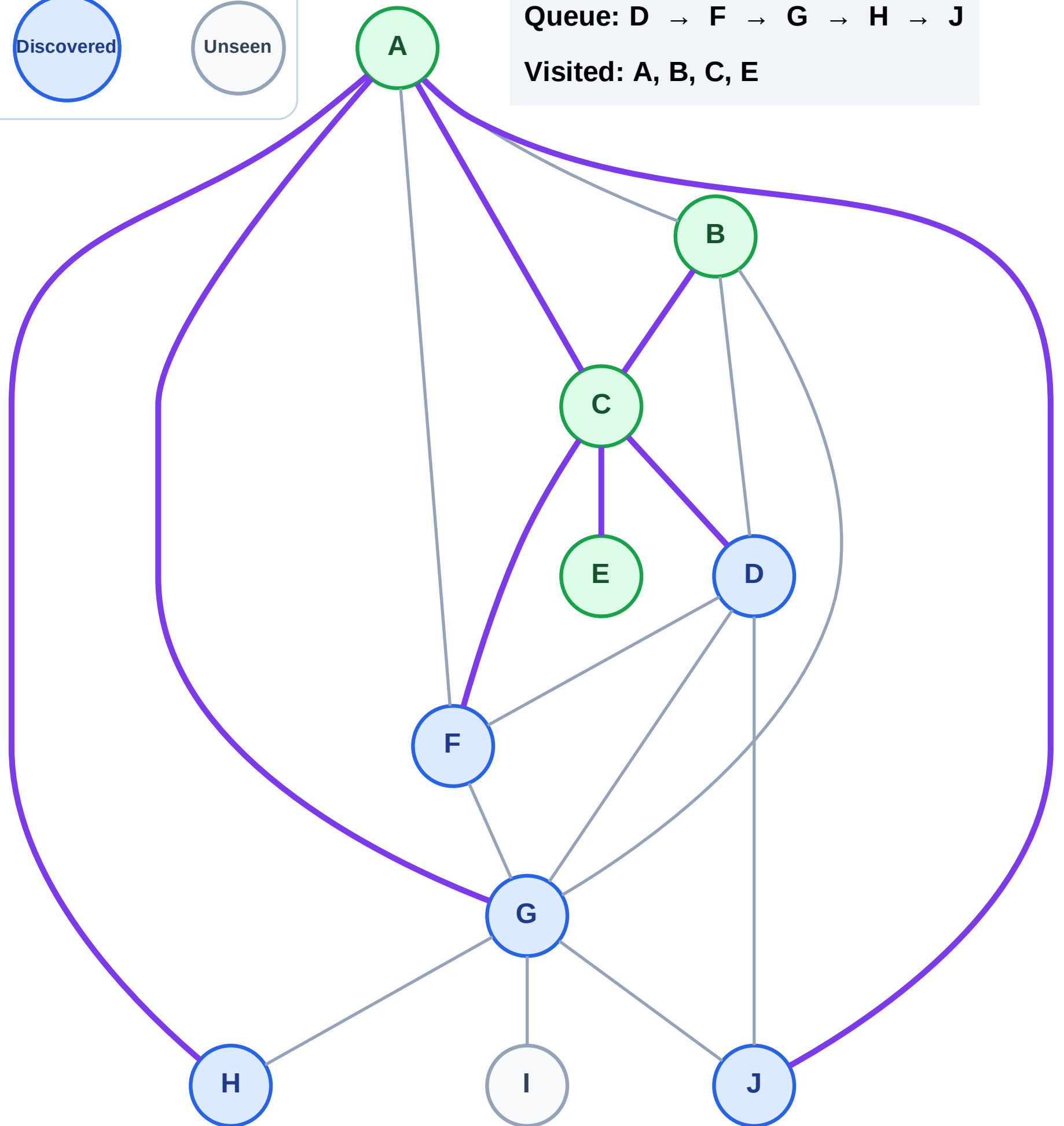
Step 9: No new neighbors from B

Legend



Queue: D → F → G → H → J

Visited: A, B, C, E



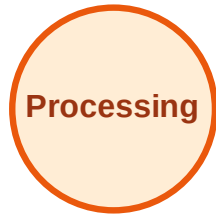
BFS Traversal

Step 10: Process node D

Legend



Complete



Processing



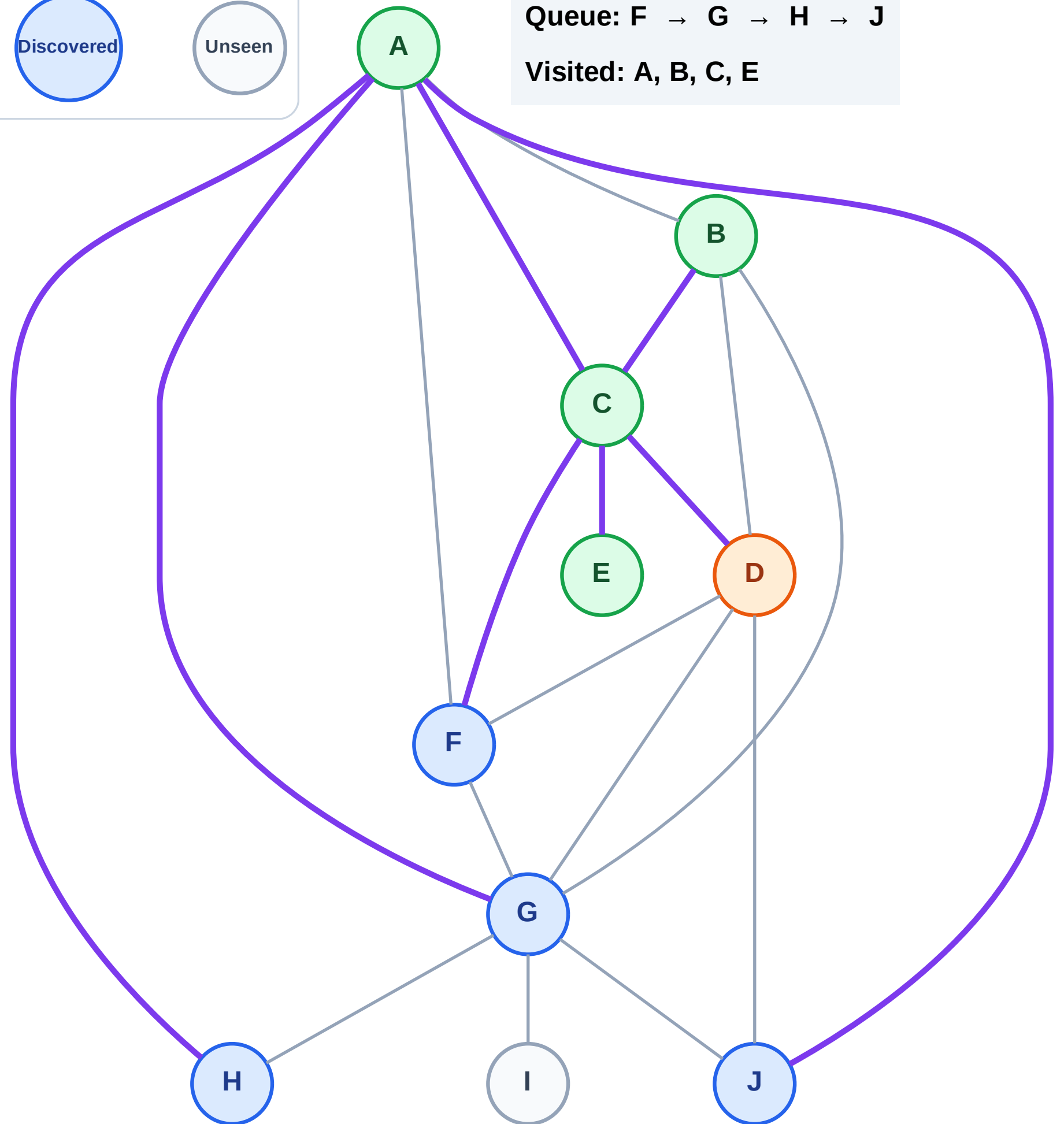
Discovered



Unseen

Queue: F → G → H → J

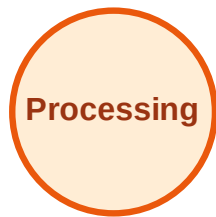
Visited: A, B, C, E



BFS Traversal

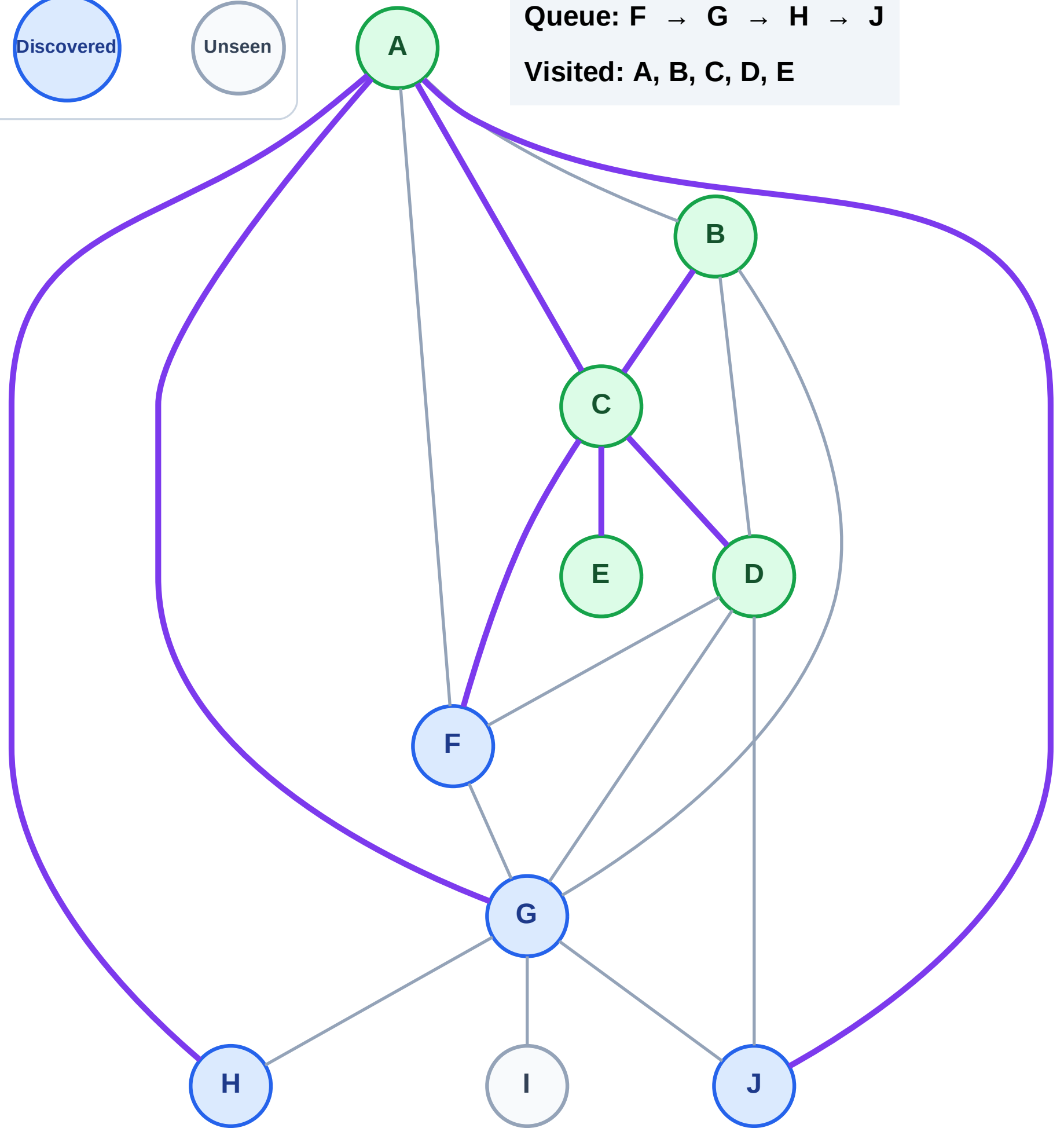
Step 11: No new neighbors from D

Legend



Queue: F → G → H → J

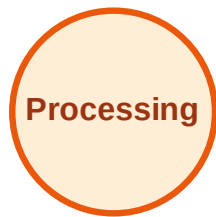
Visited: A, B, C, D, E



BFS Traversal

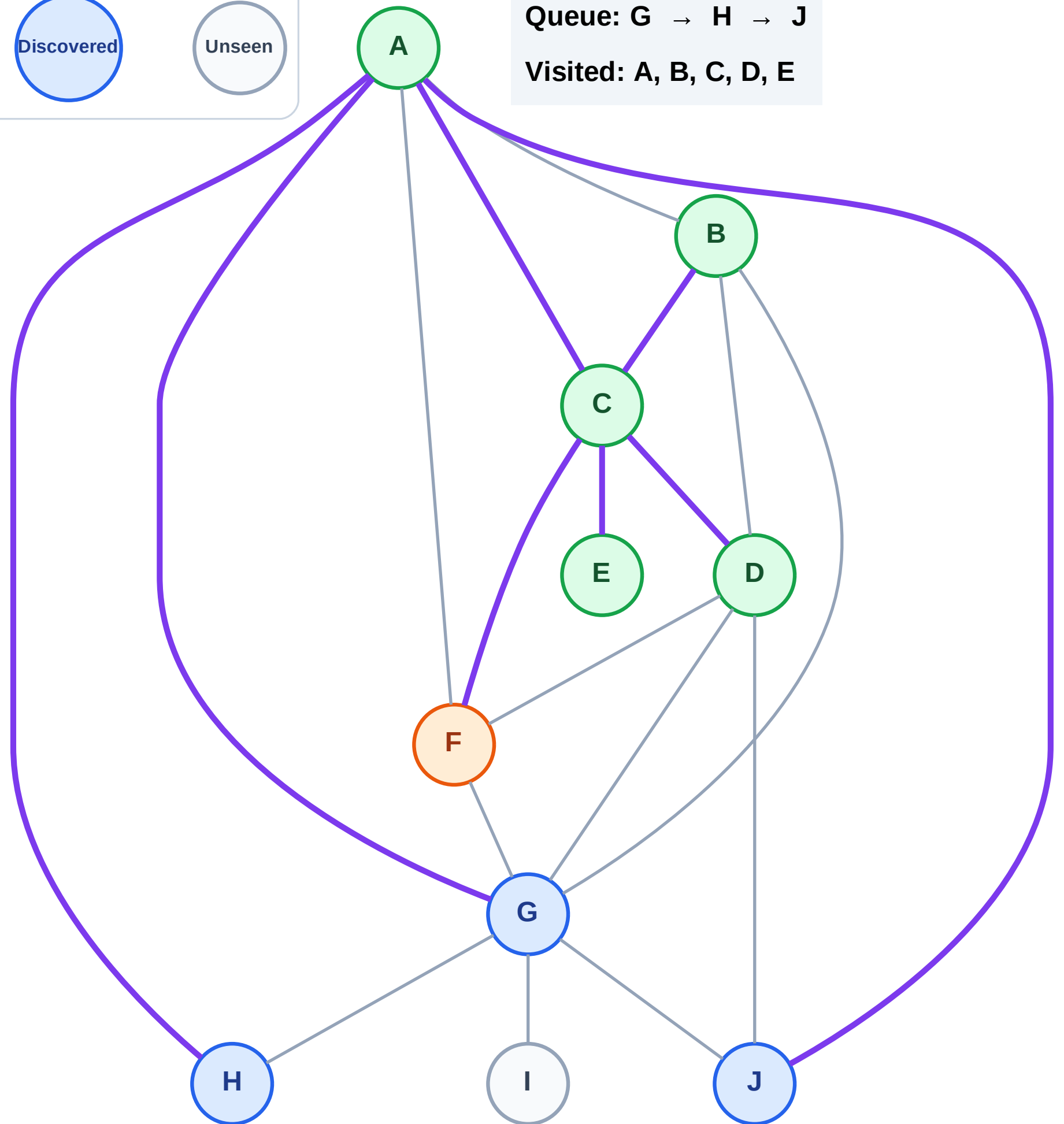
Step 12: Process node F

Legend



Queue: G → H → J

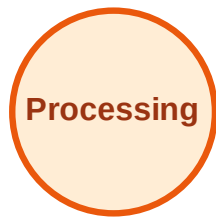
Visited: A, B, C, D, E



BFS Traversal

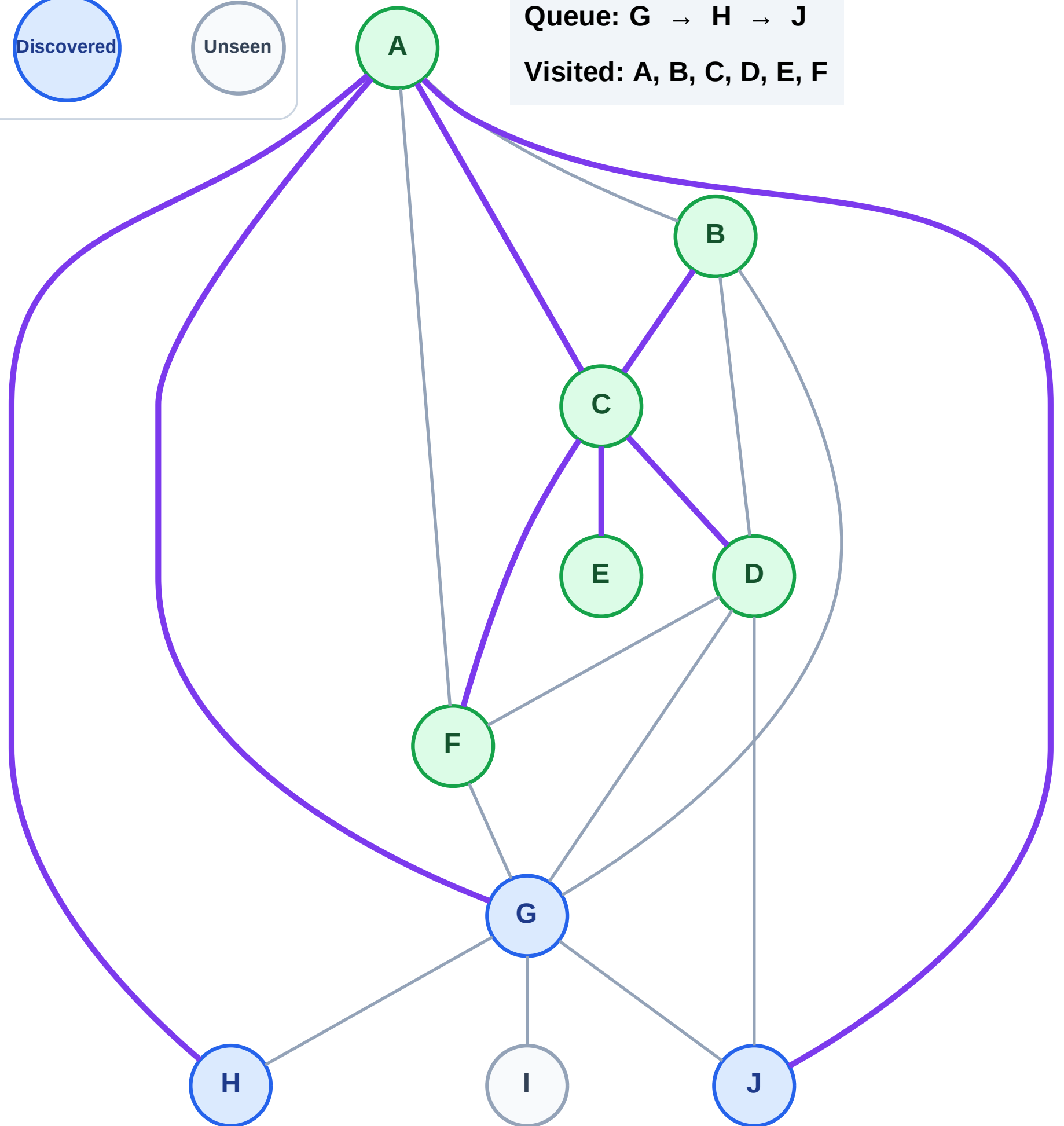
Step 13: No new neighbors from F

Legend



Queue: G → H → J

Visited: A, B, C, D, E, F



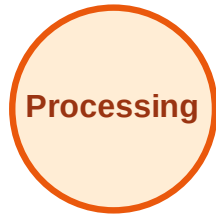
BFS Traversal

Step 14: Process node G

Legend



Complete



Processing



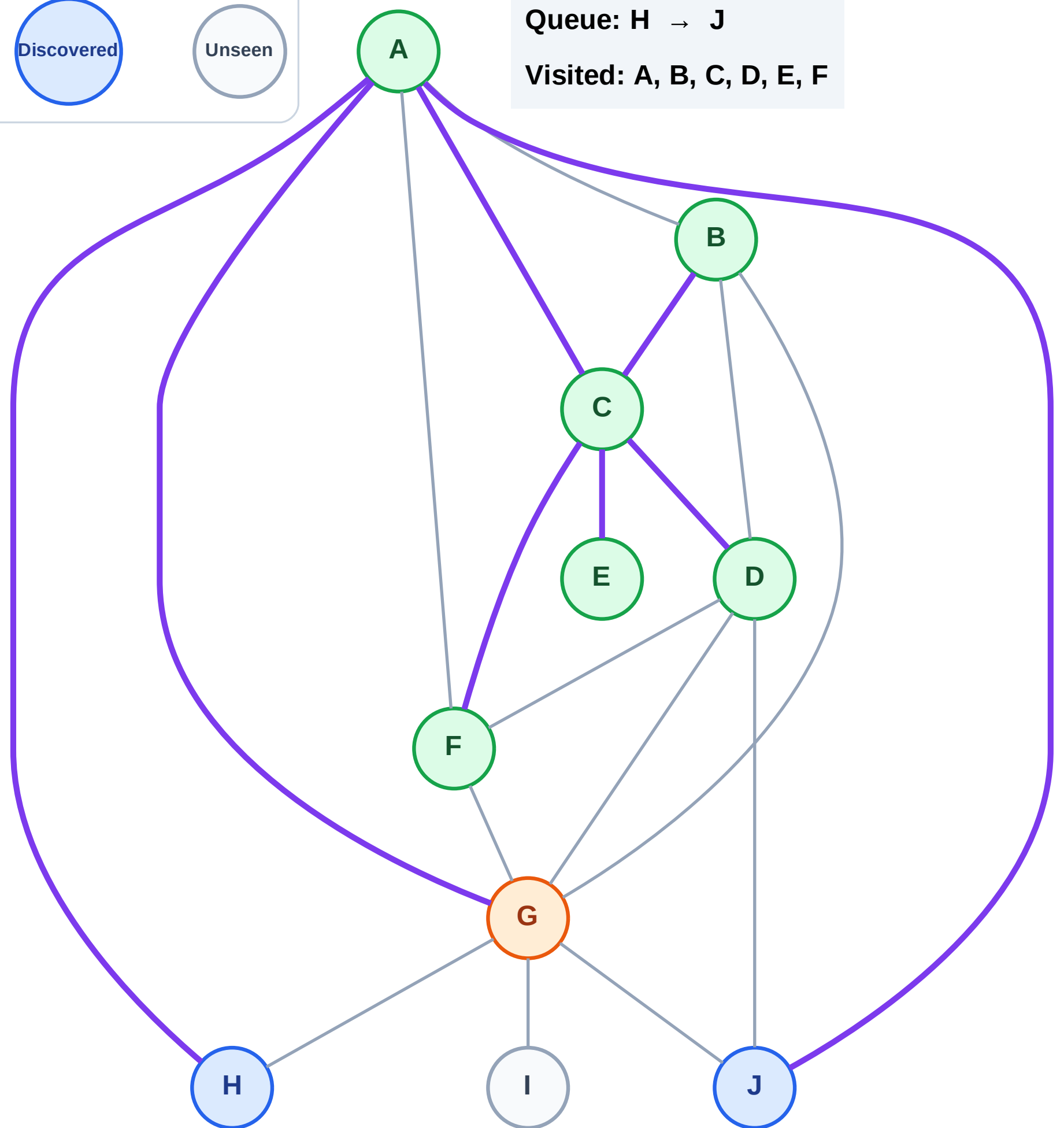
Discovered



Unseen

Queue: H → J

Visited: A, B, C, D, E, F



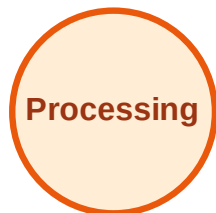
BFS Traversal

Step 15: Discover I from G

Legend



Complete



Processing



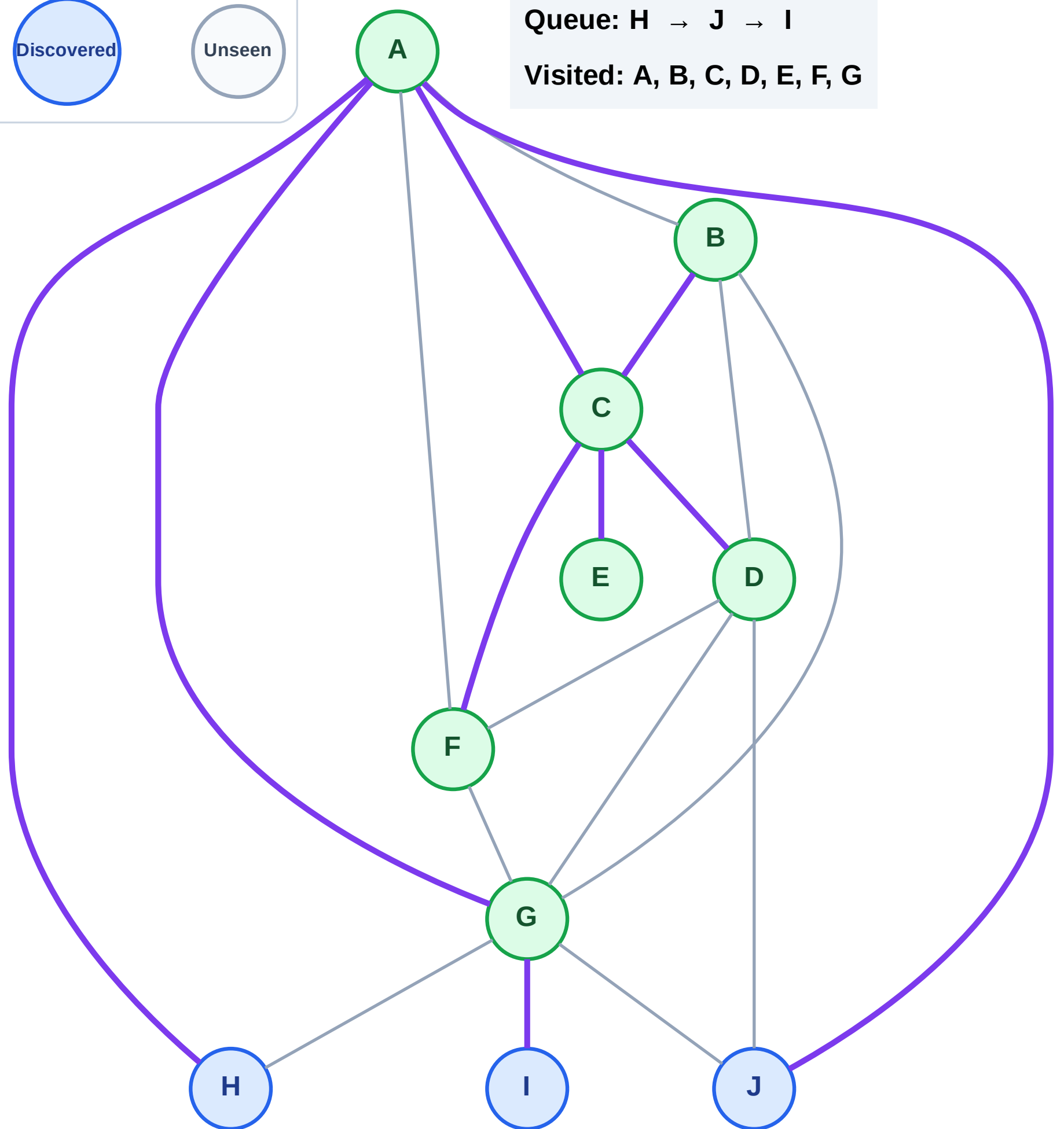
Discovered



Unseen

Queue: H → J → I

Visited: A, B, C, D, E, F, G



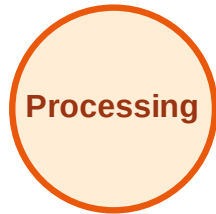
BFS Traversal

Step 16: Process node H

Legend



Complete



Processing



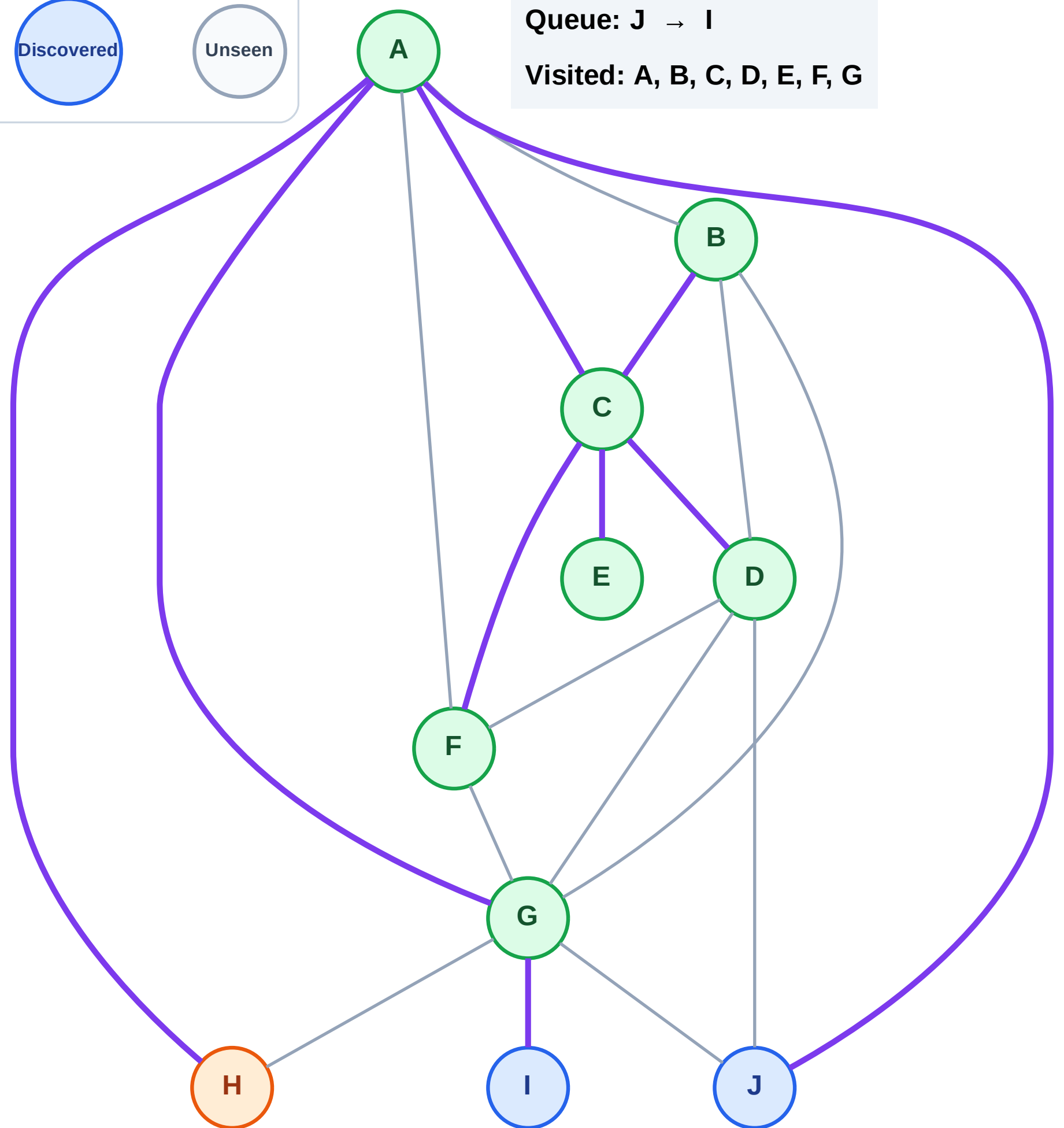
Discovered



Unseen

Queue: J → I

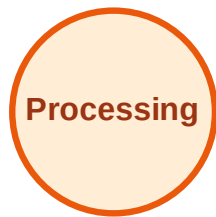
Visited: A, B, C, D, E, F, G



BFS Traversal

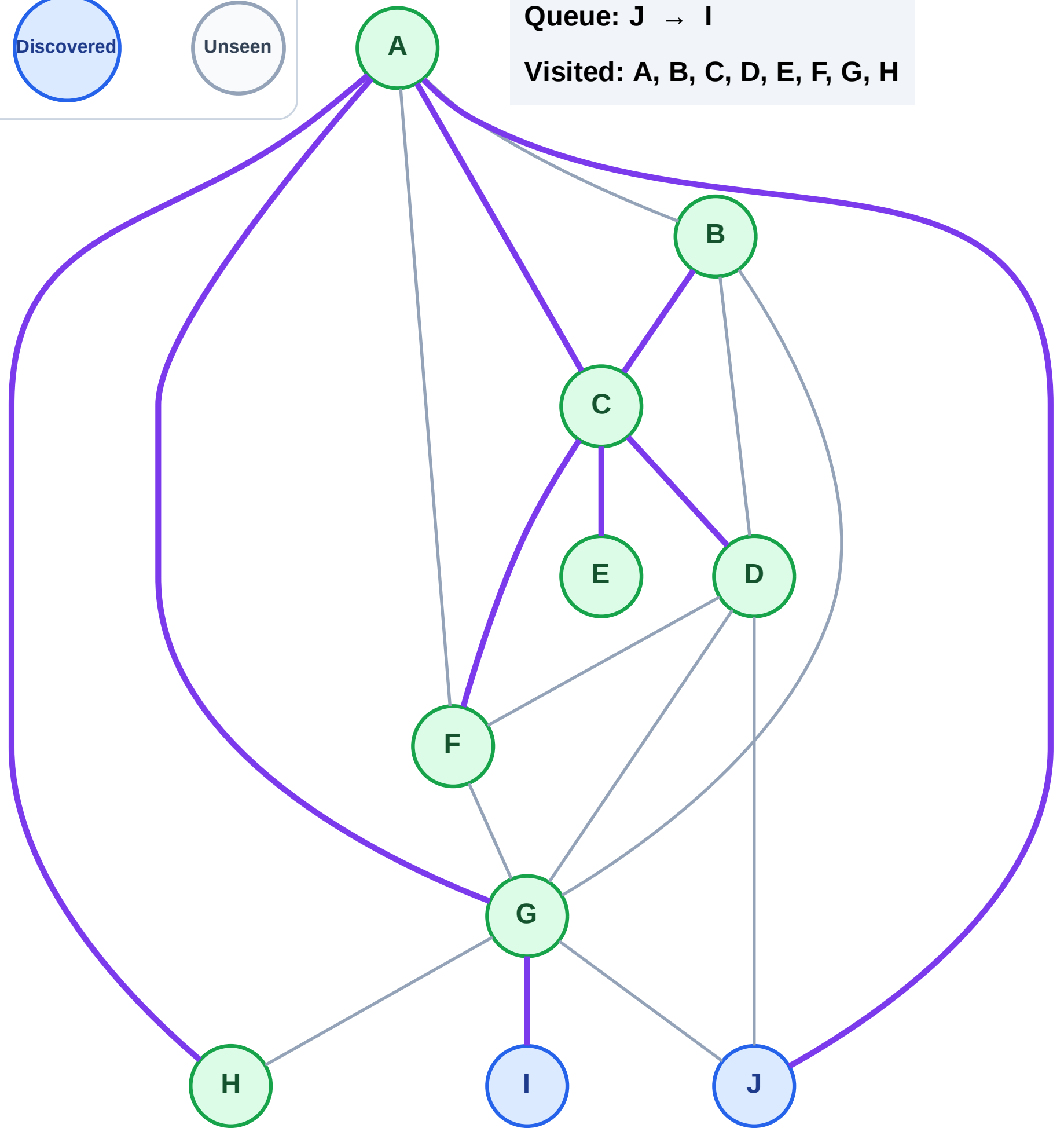
Step 17: No new neighbors from H

Legend



Queue: J → I

Visited: A, B, C, D, E, F, G, H



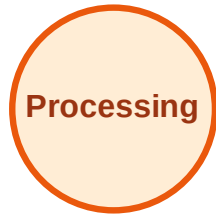
BFS Traversal

Step 18: Process node J

Legend



Complete



Processing



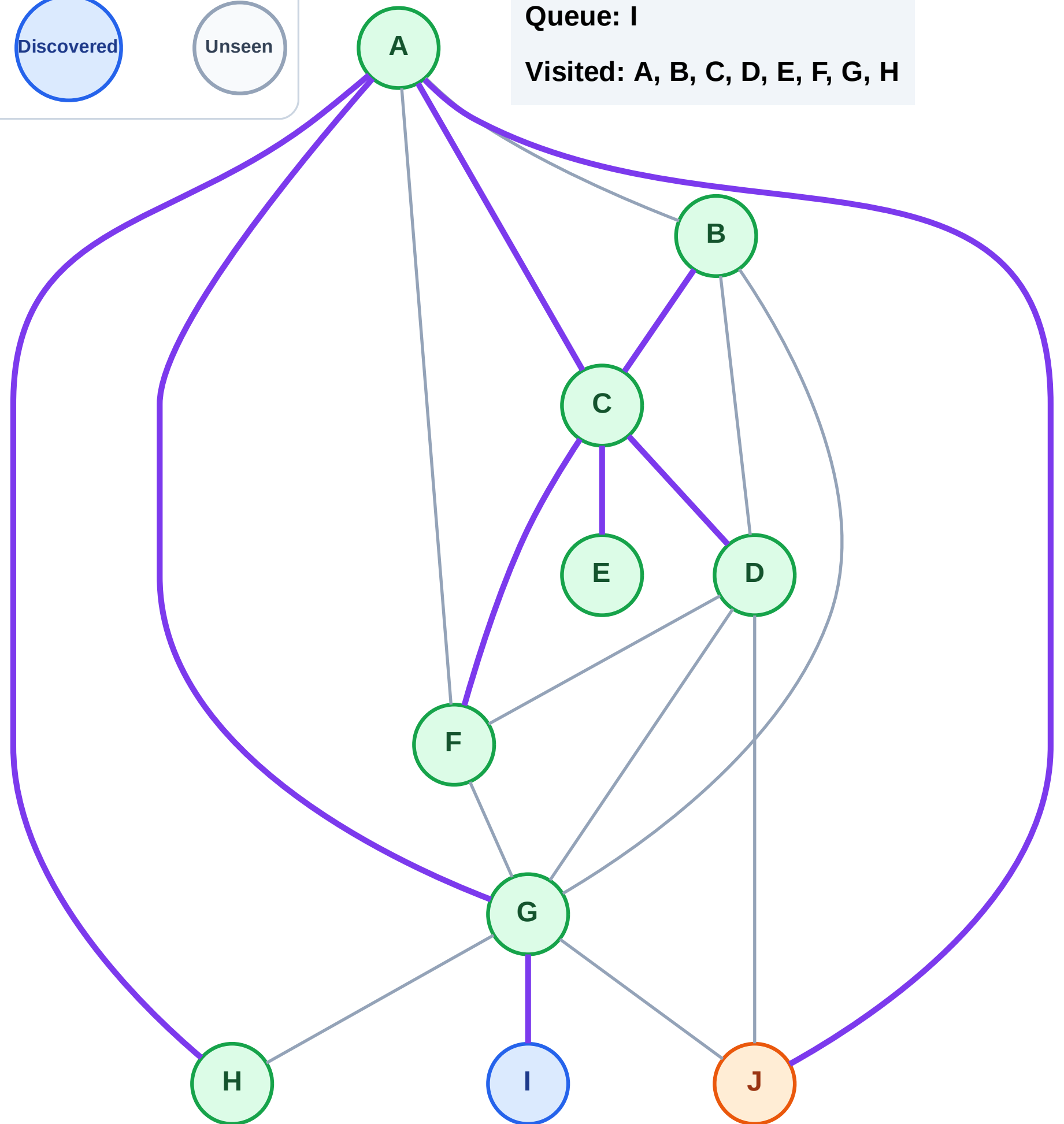
Discovered



Unseen

Queue: I

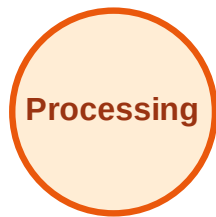
Visited: A, B, C, D, E, F, G, H



BFS Traversal

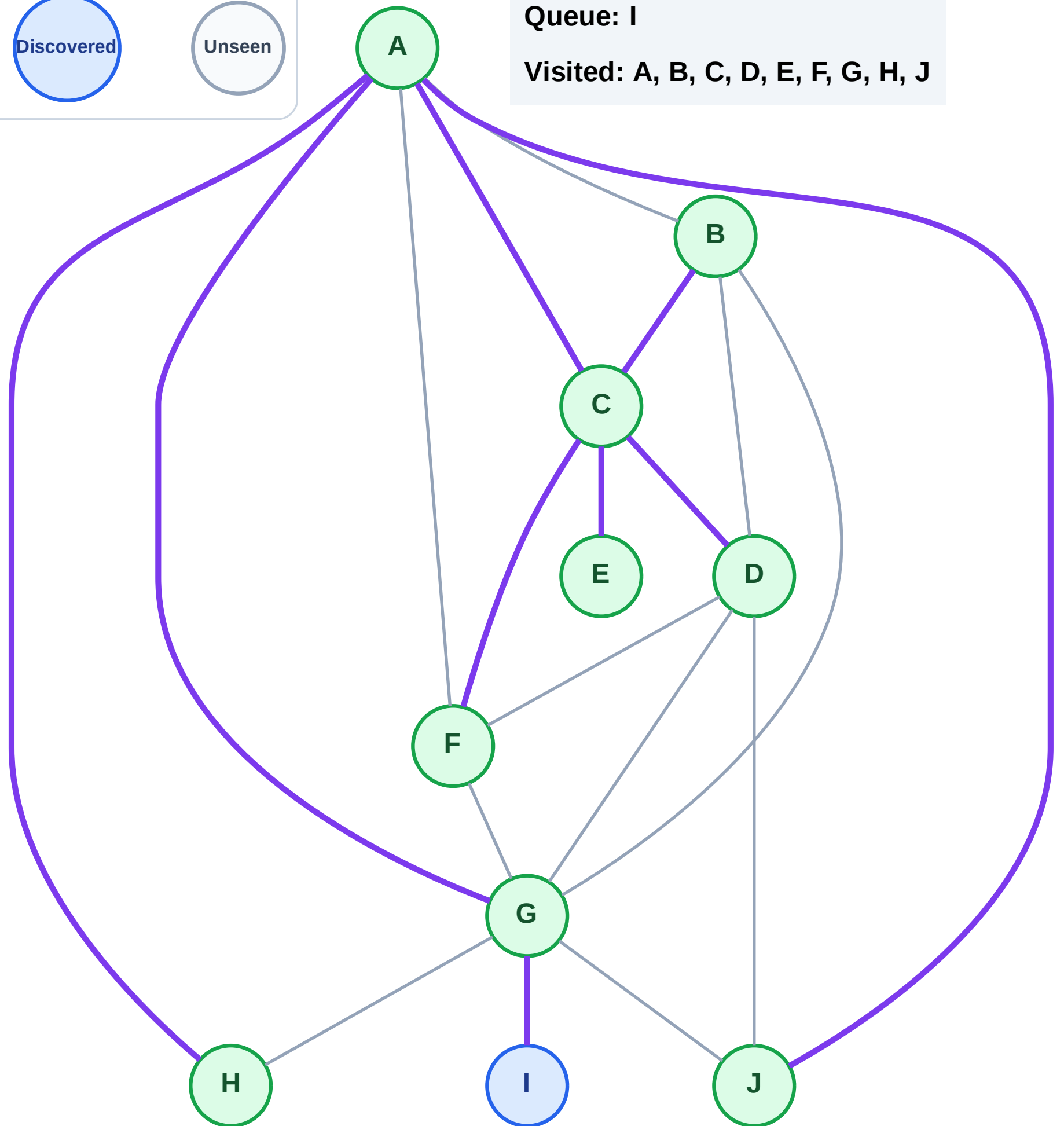
Step 19: No new neighbors from J

Legend



Queue: I

Visited: A, B, C, D, E, F, G, H, J



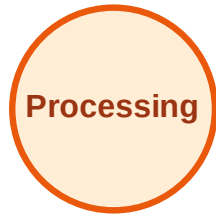
BFS Traversal

Step 20: Process node I

Legend



Complete



Processing



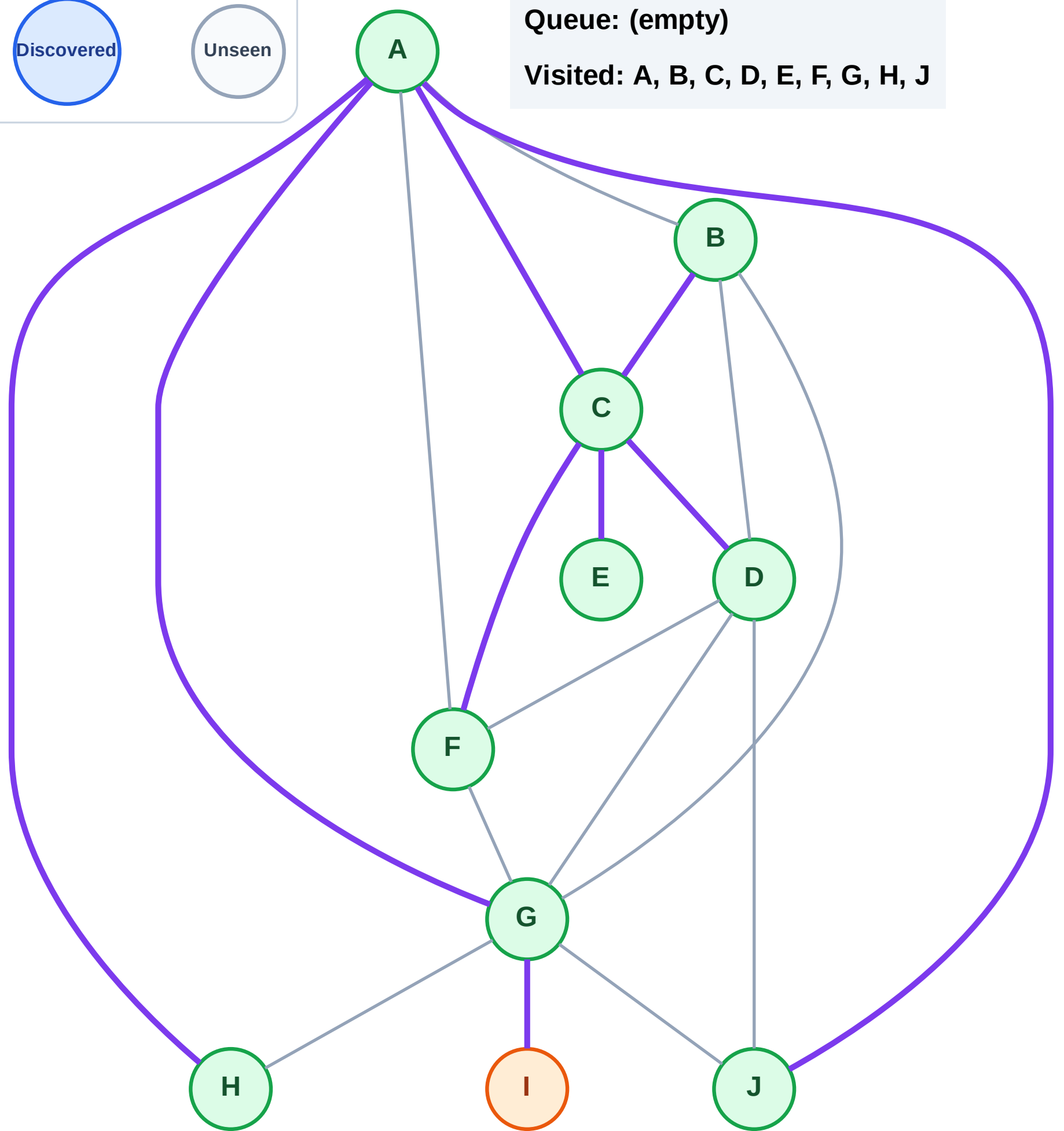
Discovered



Unseen

Queue: (empty)

Visited: A, B, C, D, E, F, G, H, J



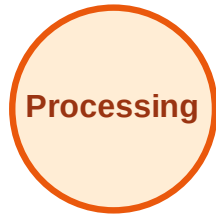
BFS Traversal

Step 21: No new neighbors from I

Legend



Complete



Processing



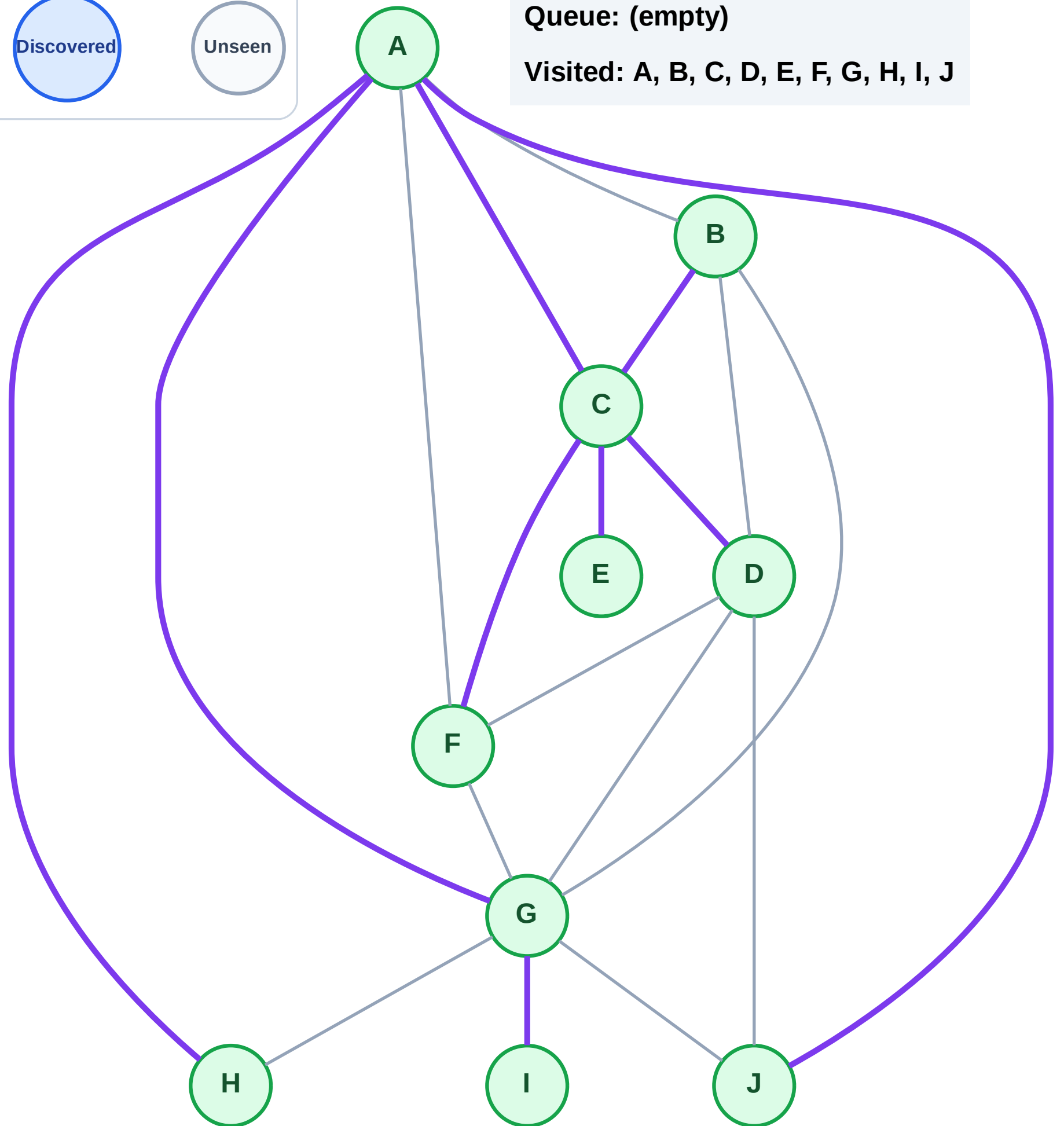
Discovered



Unseen

Queue: (empty)

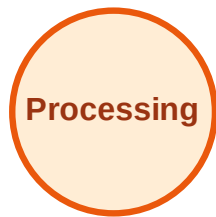
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

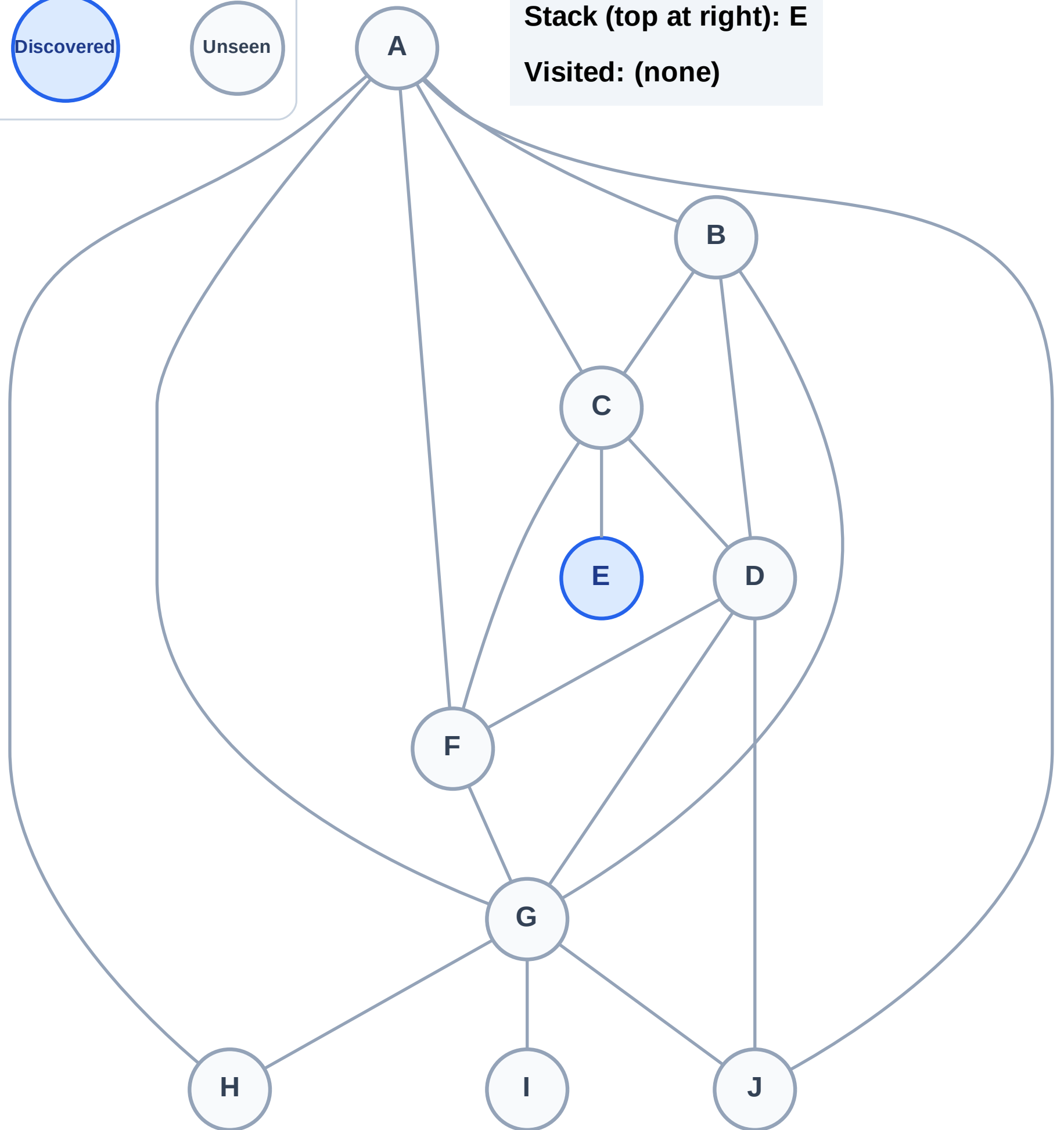
Step 1: Start at E

Legend



Stack (top at right): E

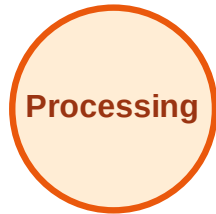
Visited: (none)



DFS Traversal

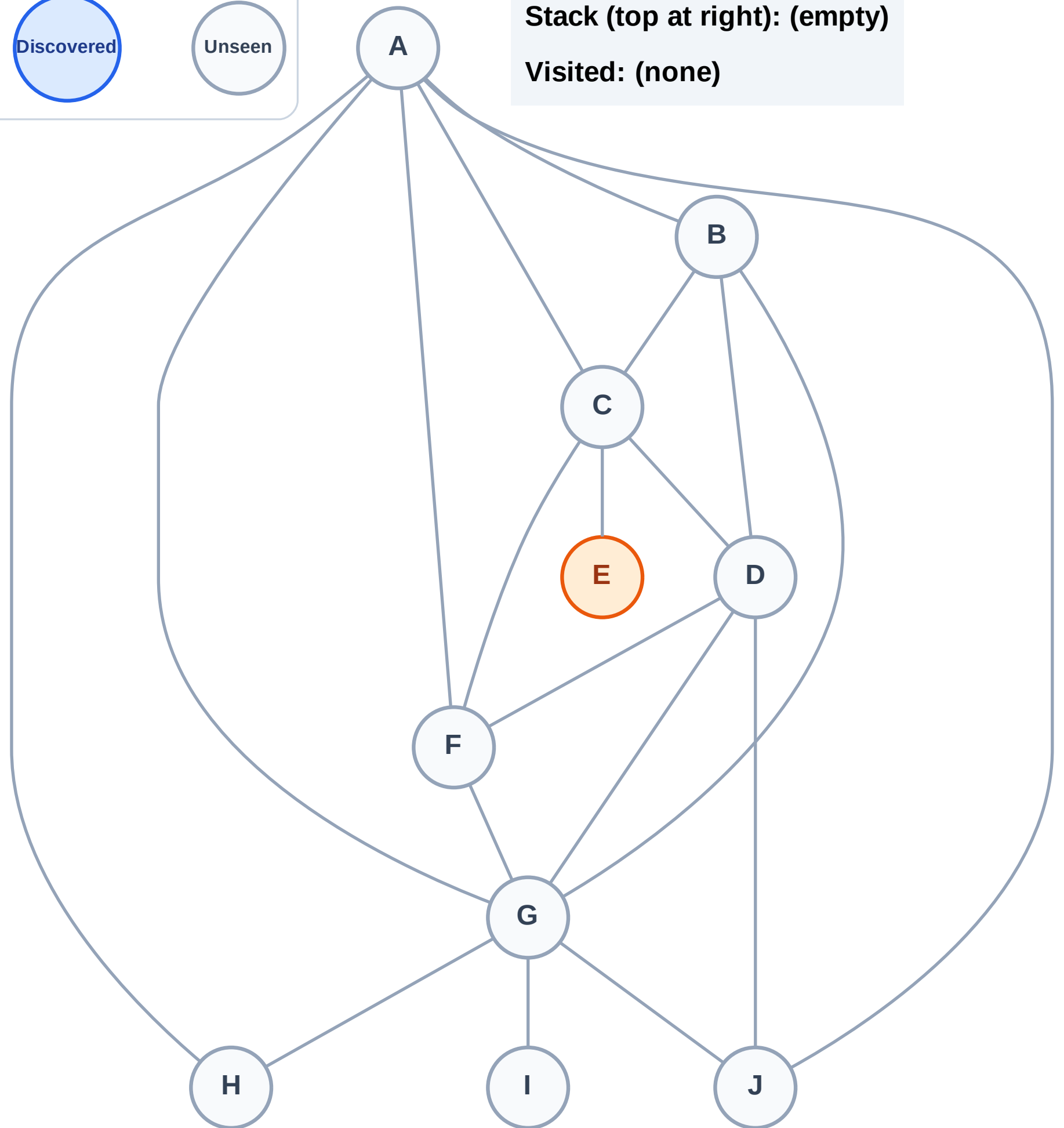
Step 2: Process node E

Legend



Stack (top at right): (empty)

Visited: (none)



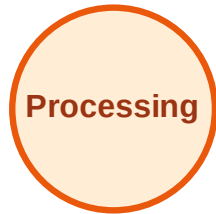
DFS Traversal

Step 3: Discover C from E

Legend



Complete



Processing



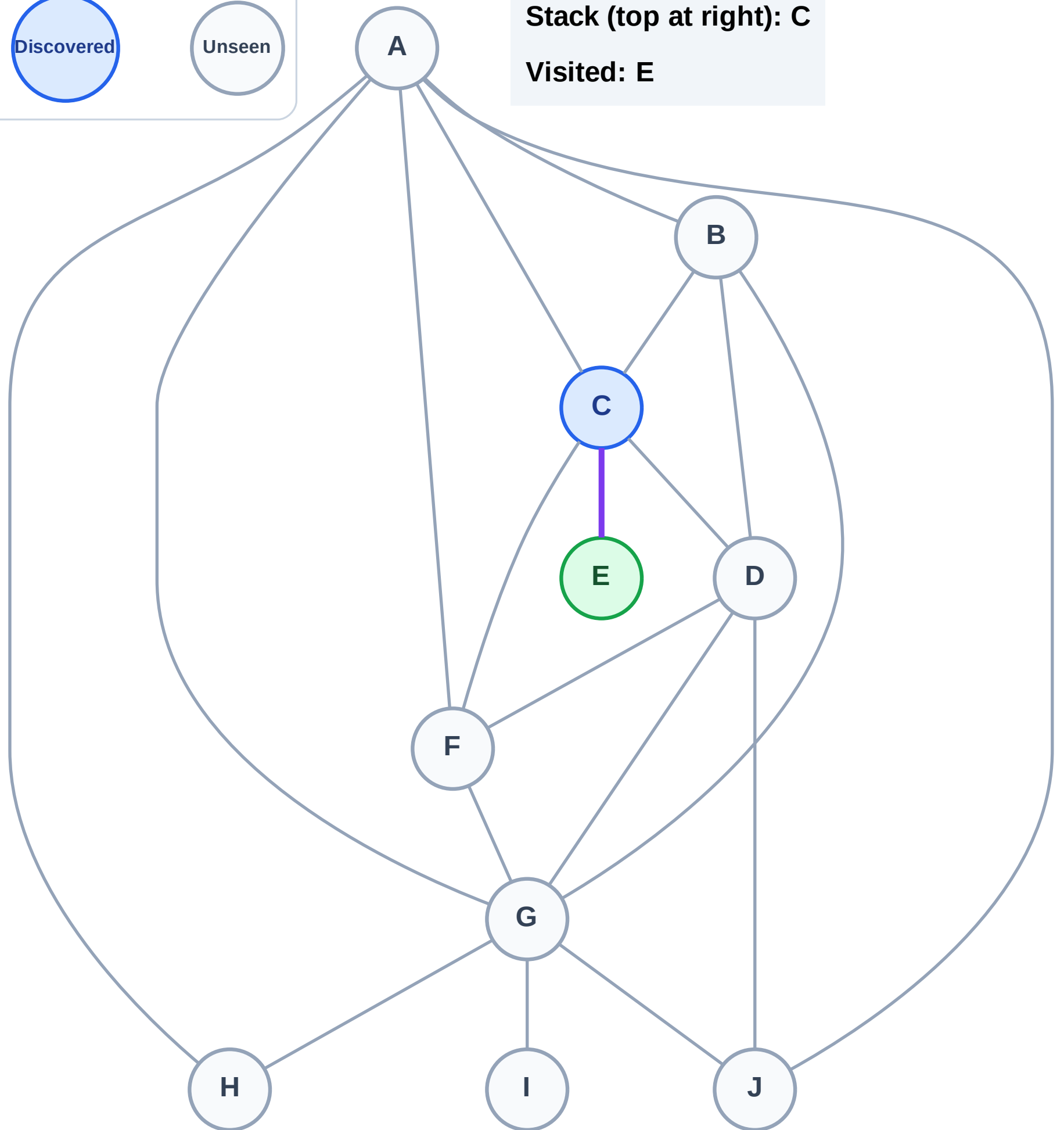
Discovered



Unseen

Stack (top at right): C

Visited: E



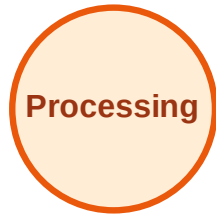
DFS Traversal

Step 4: Process node C

Legend



Complete



Processing



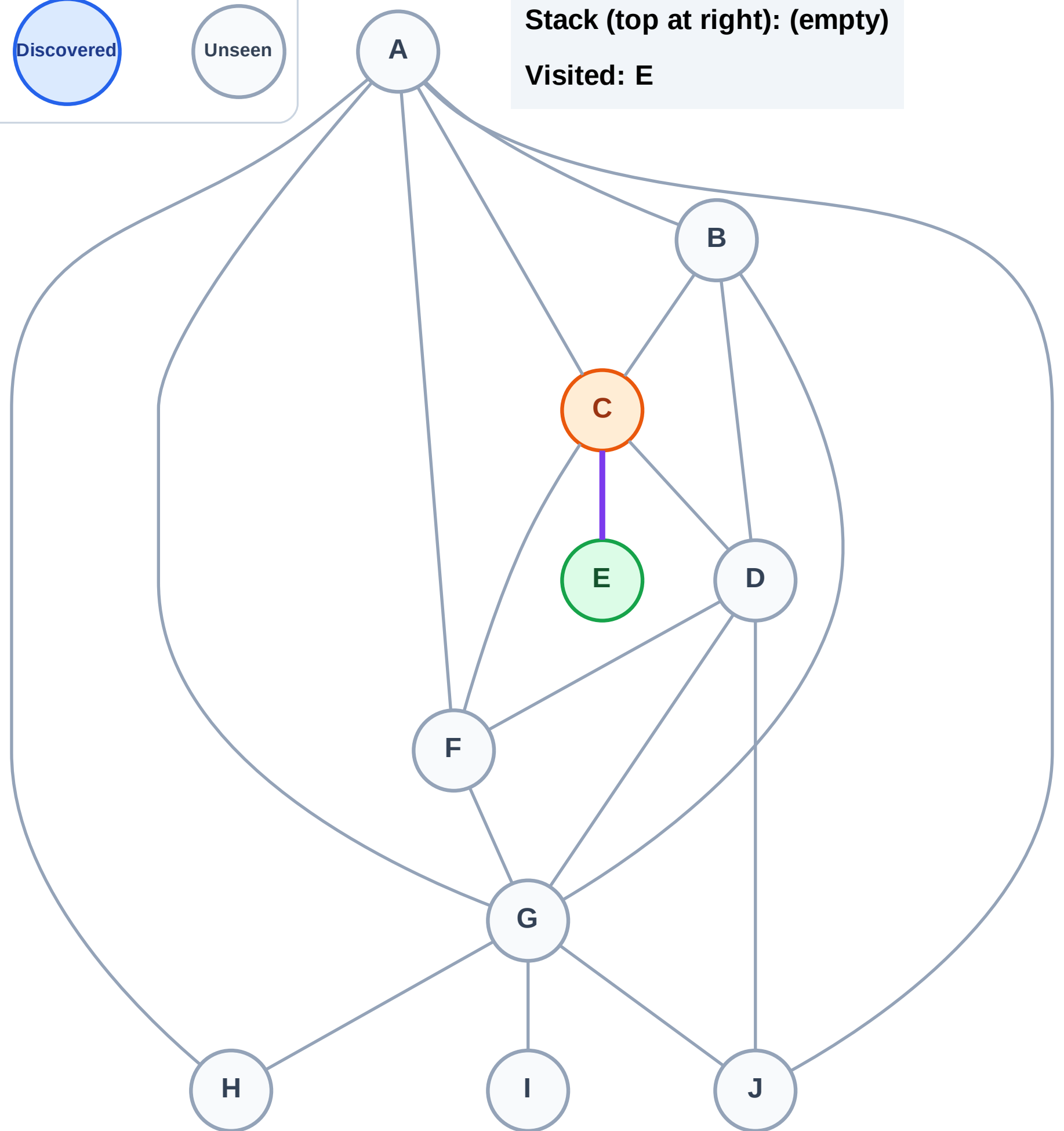
Discovered



Unseen

Stack (top at right): (empty)

Visited: E



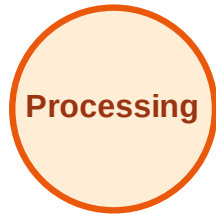
DFS Traversal

Step 5: Discover F, D, B, A from C

Legend



Complete



Processing



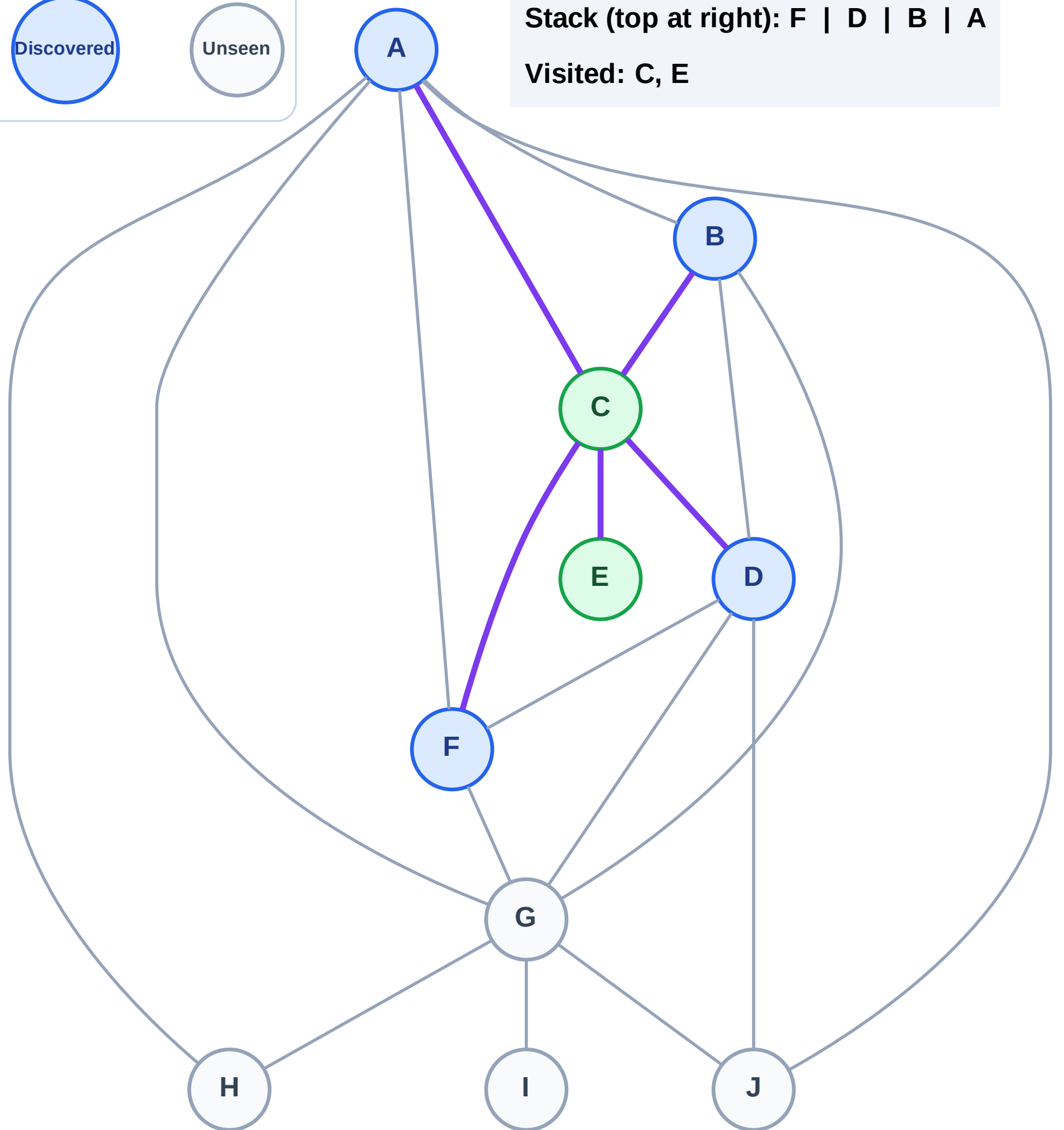
Discovered



Unseen

Stack (top at right): F | D | B | A

Visited: C, E



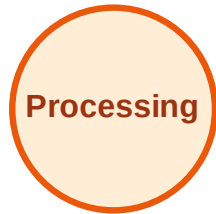
DFS Traversal

Step 6: Process node A

Legend



Complete



Processing



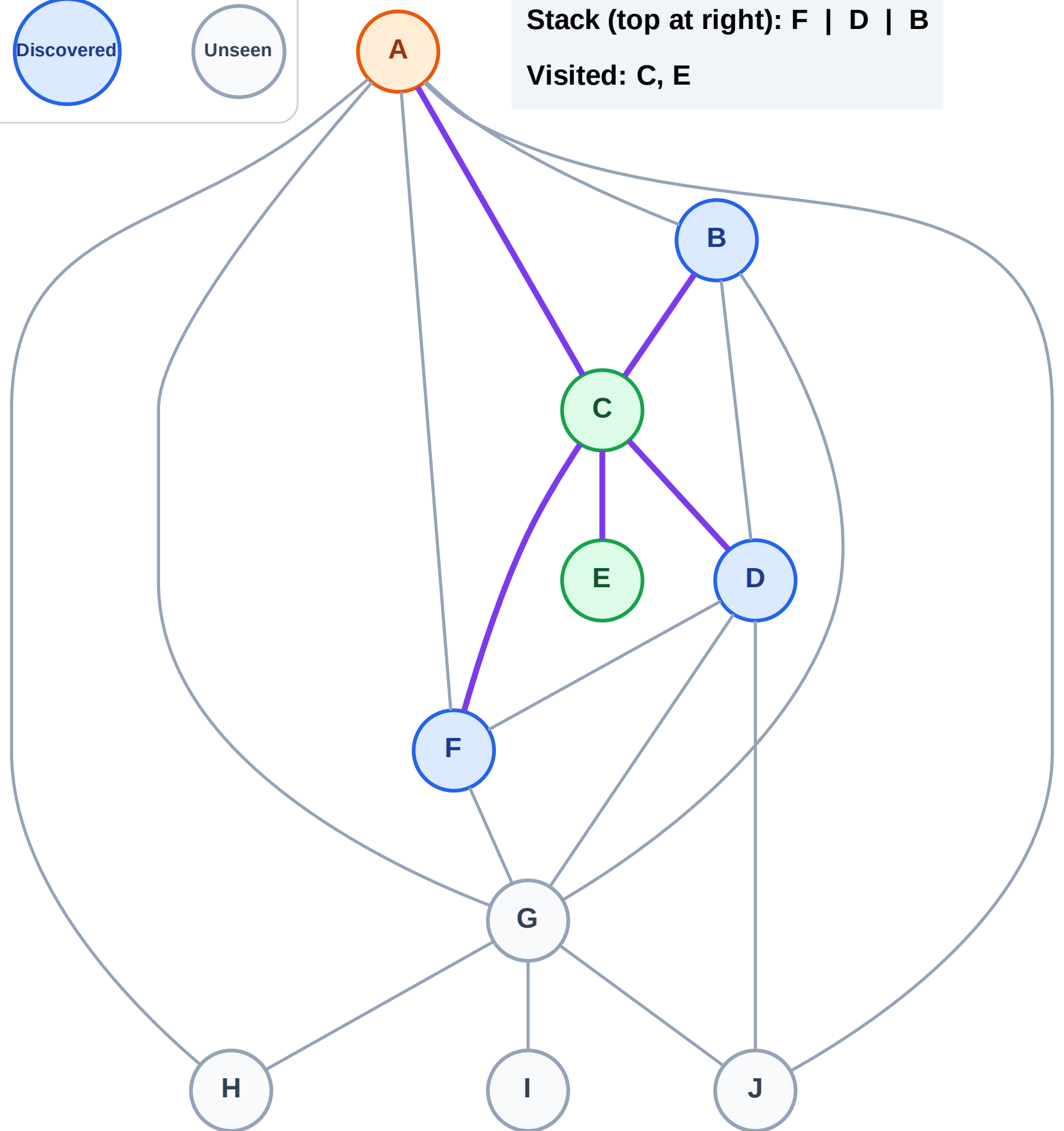
Discovered



Unseen

Stack (top at right): F | D | B

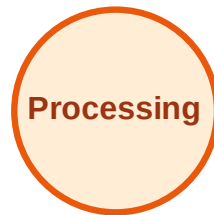
Visited: C, E



DFS Traversal

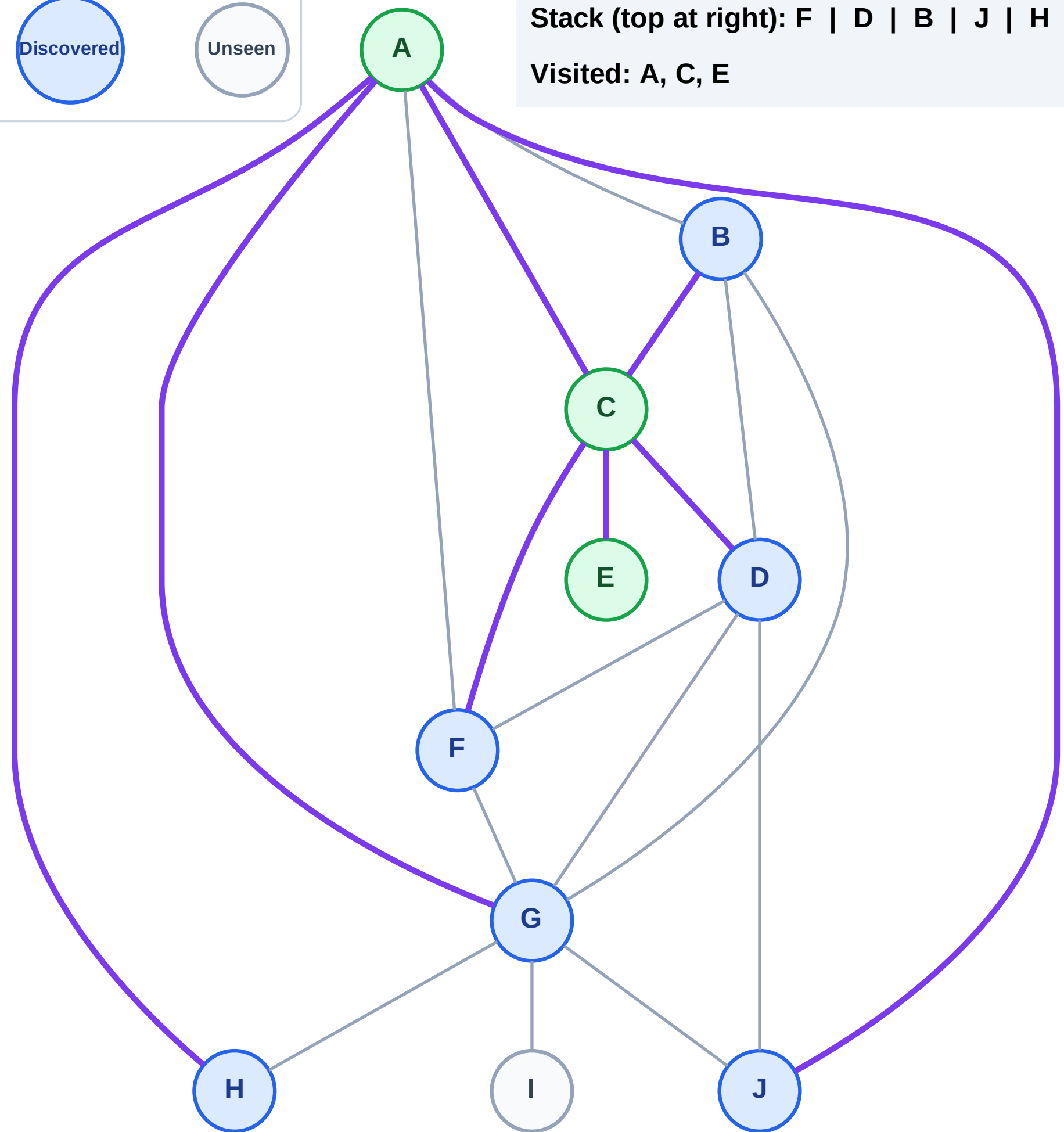
Step 7: Discover J, H, G from A

Legend



Stack (top at right): F | D | B | J | H | G

Visited: A, C, E



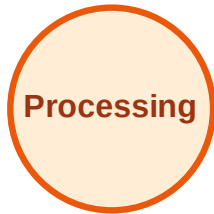
DFS Traversal

Step 8: Process node G

Legend



Complete



Processing



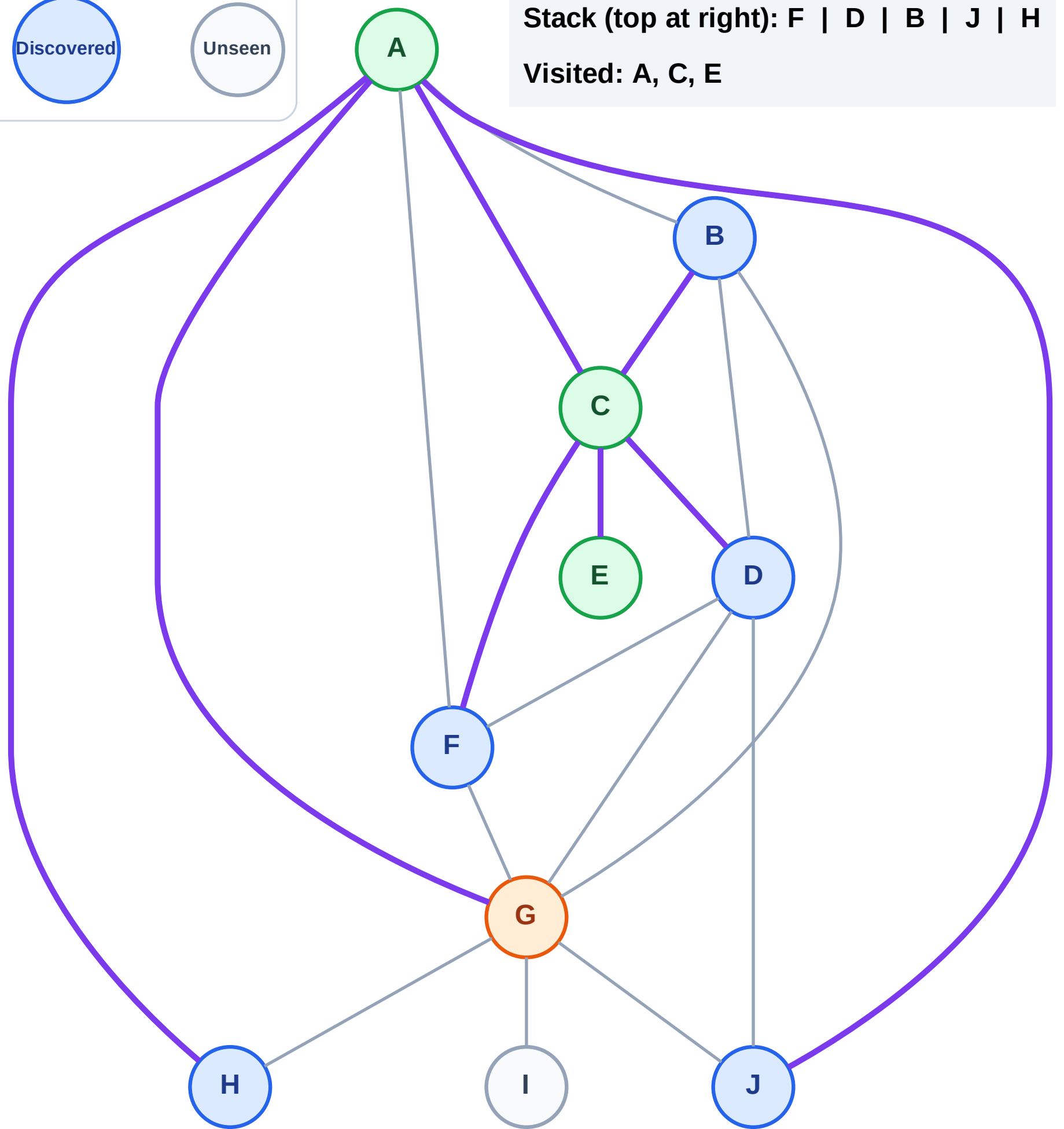
Discovered



Unseen

Stack (top at right): F | D | B | J | H

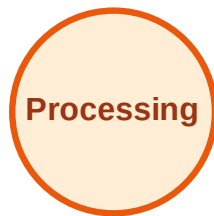
Visited: A, C, E



DFS Traversal

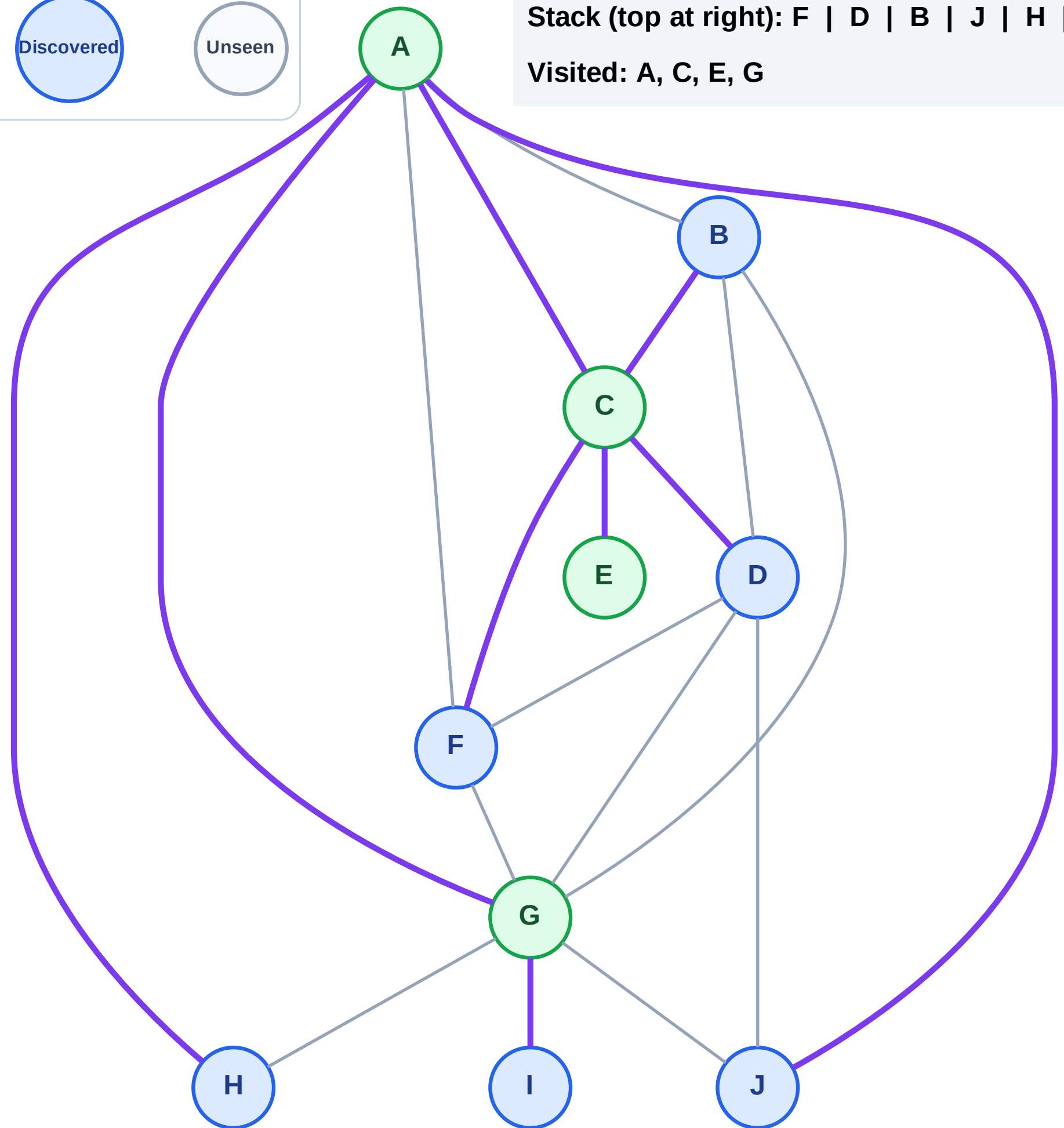
Step 9: Discover I from G

Legend



Stack (top at right): F | D | B | J | H | I

Visited: A, C, E, G



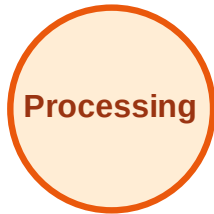
DFS Traversal

Step 10: Process node I

Legend



Complete



Processing



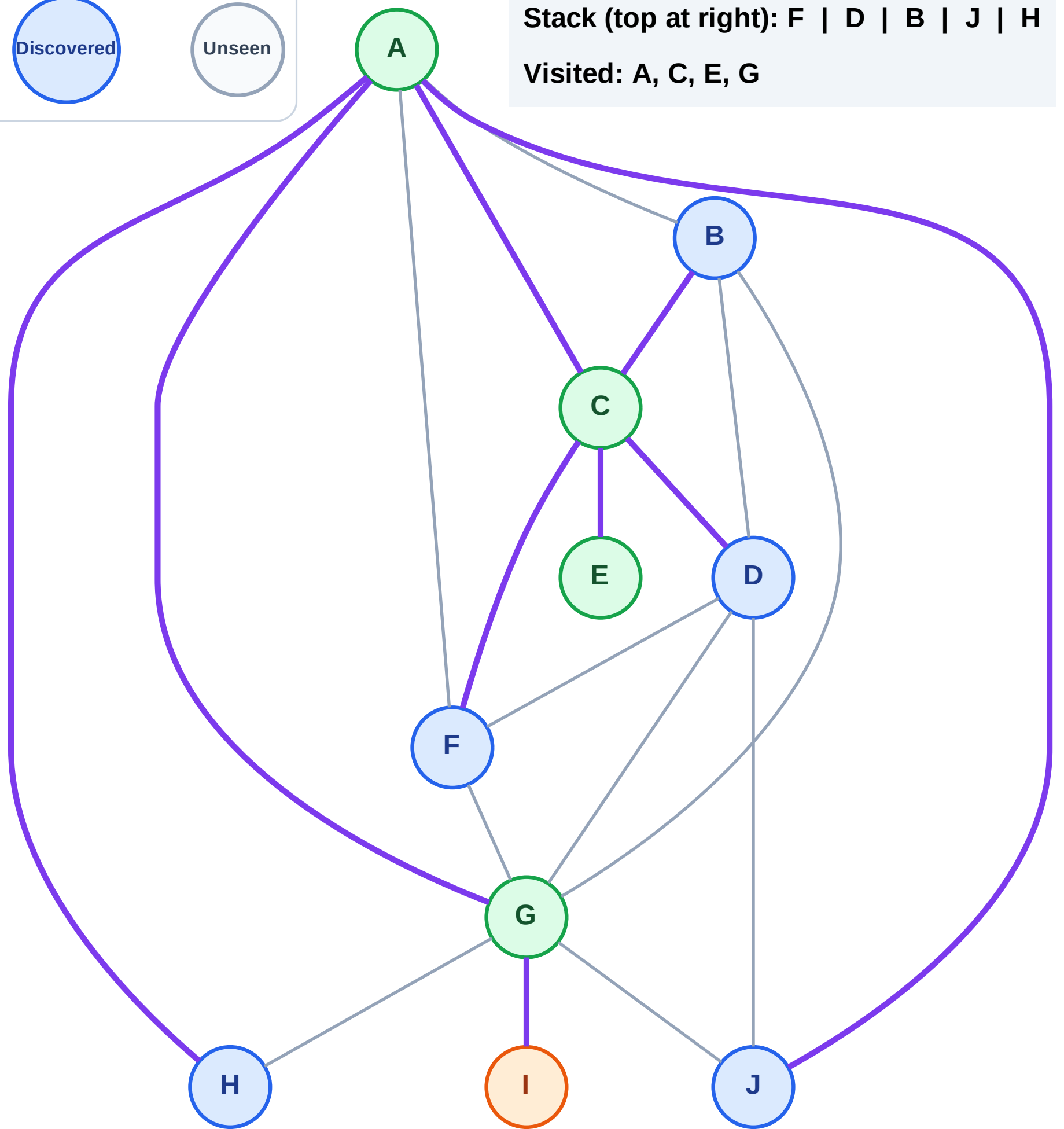
Discovered



Unseen

Stack (top at right): F | D | B | J | H

Visited: A, C, E, G

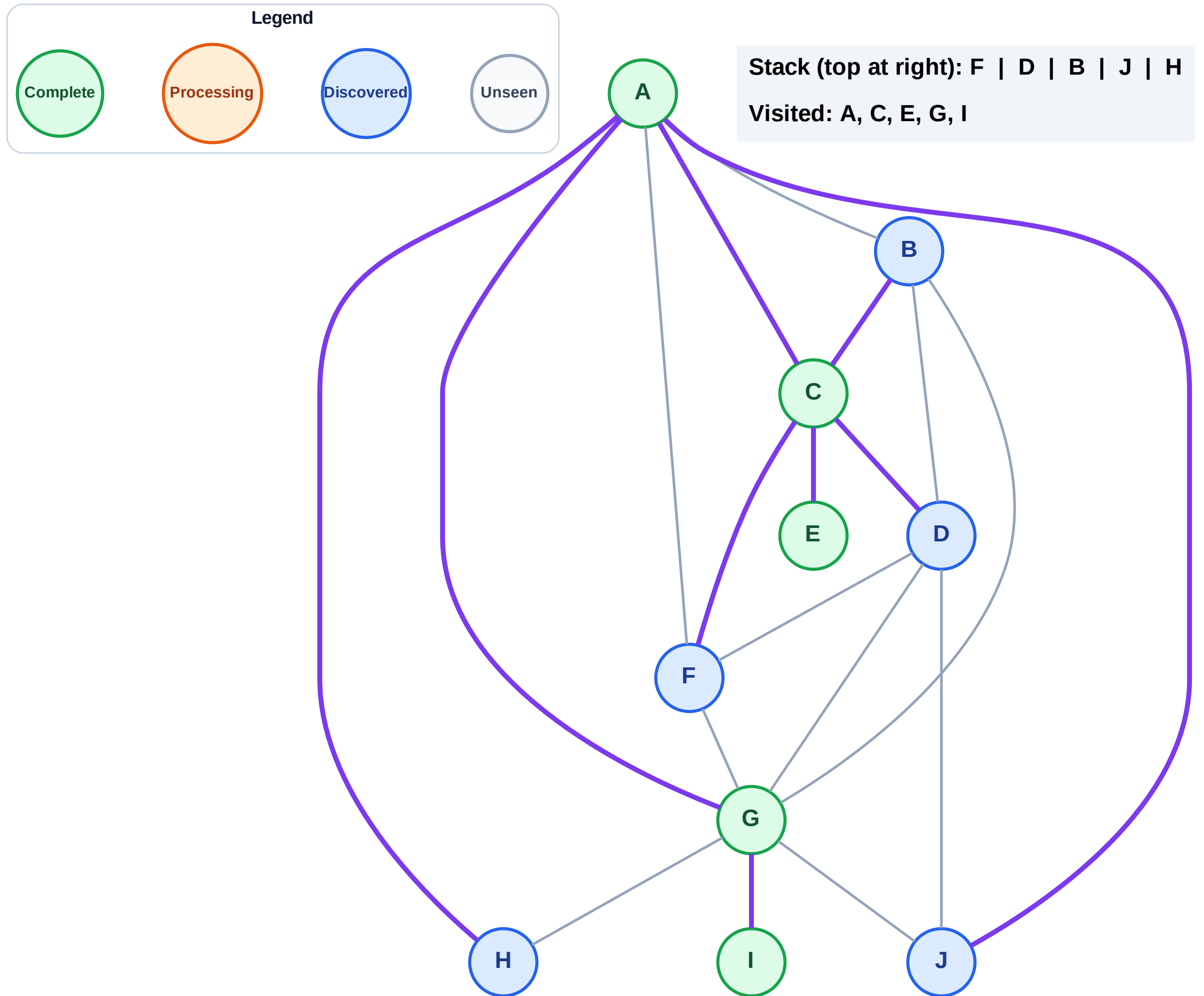


DFS Traversal

Step 11: No new neighbors from I

DFS Traversal

Step 11: No new neighbors from I



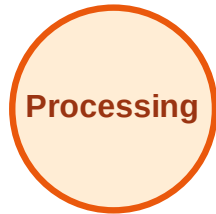
DFS Traversal

Step 12: Process node H

Legend



Complete



Processing



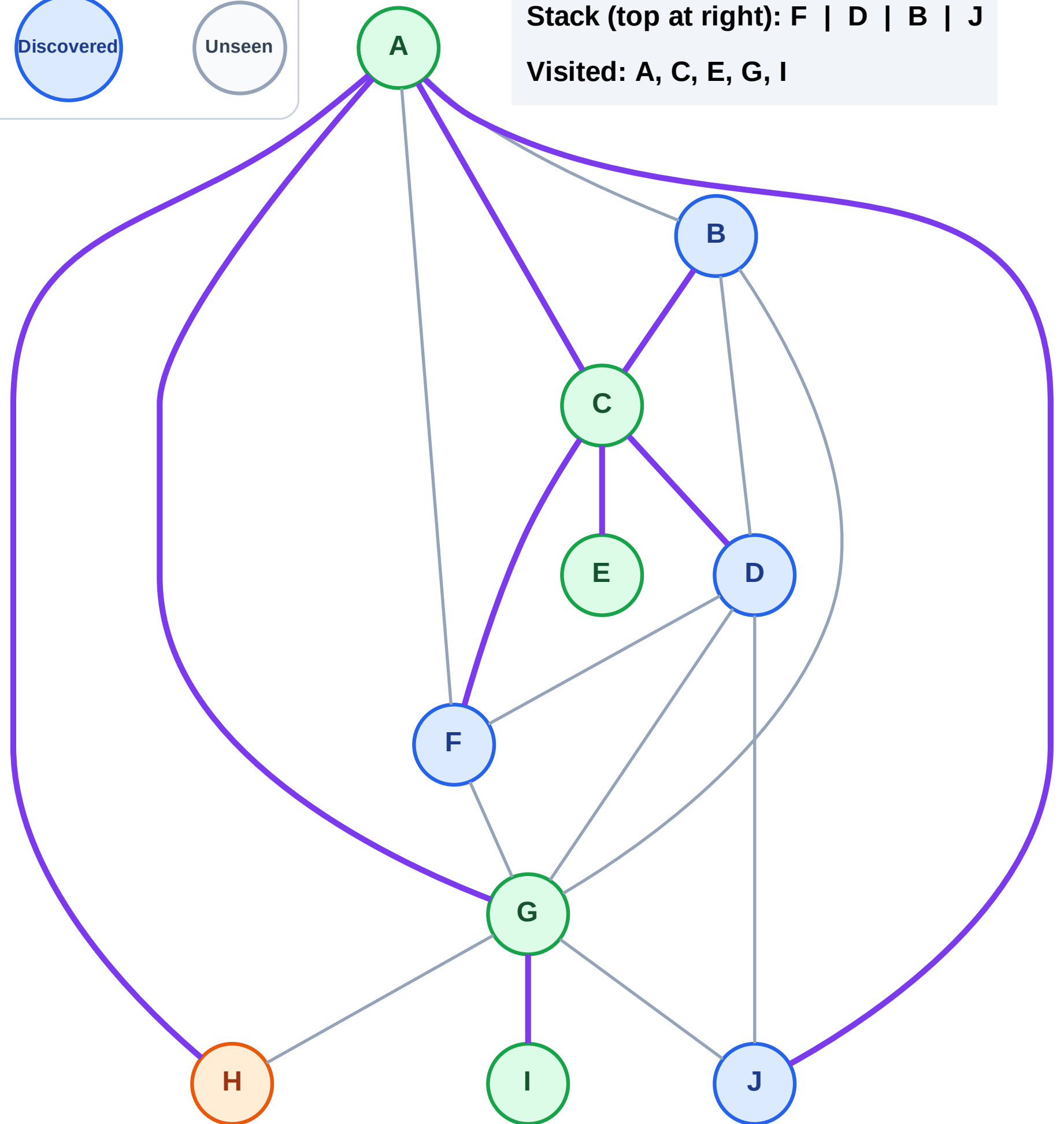
Discovered



Unseen

Stack (top at right): F | D | B | J

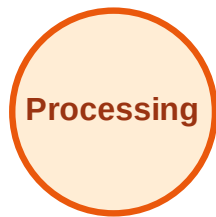
Visited: A, C, E, G, I



DFS Traversal

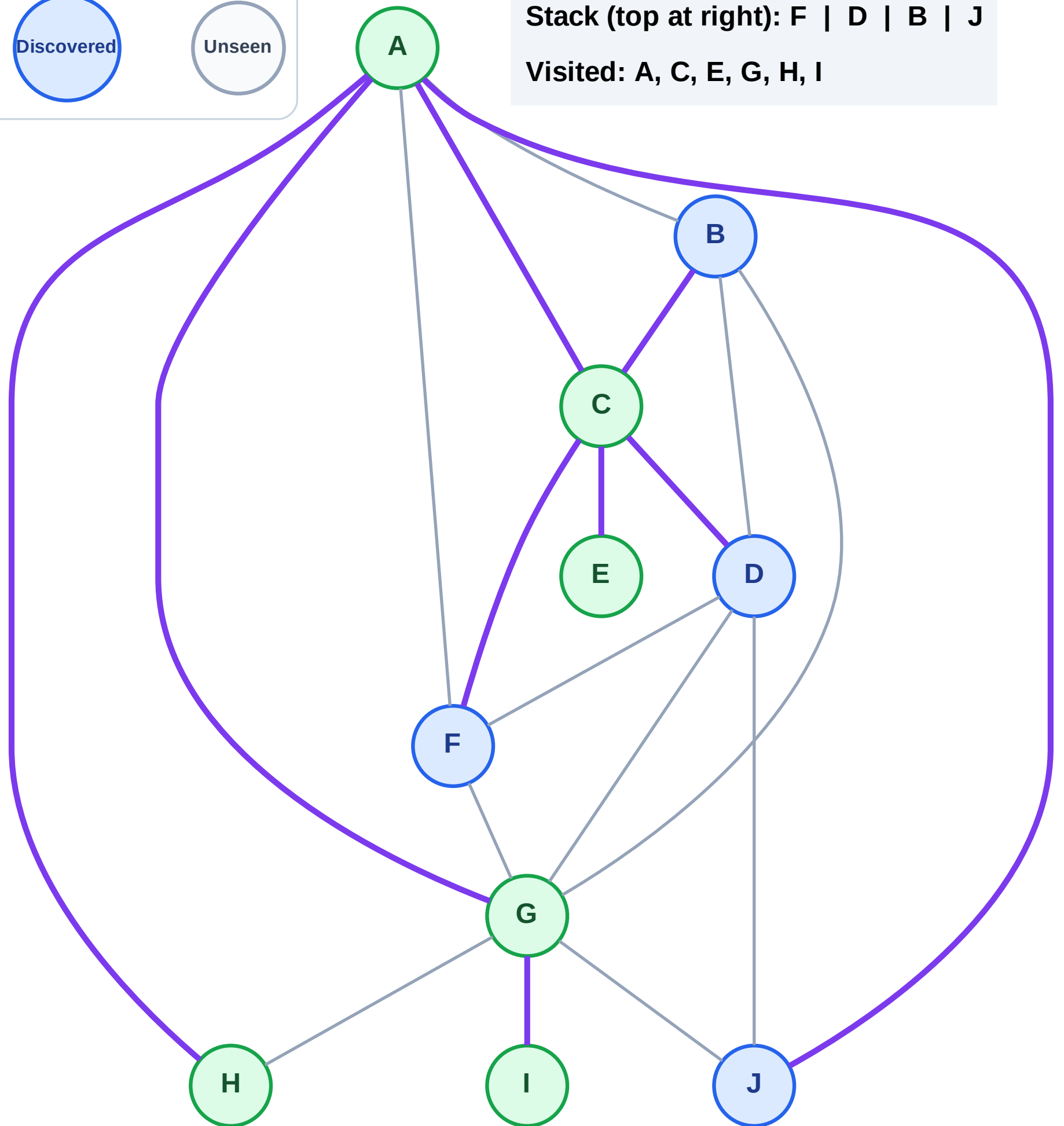
Step 13: No new neighbors from H

Legend



Stack (top at right): F | D | B | J

Visited: A, C, E, G, H, I



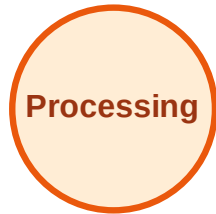
DFS Traversal

Step 14: Process node J

Legend



Complete



Processing



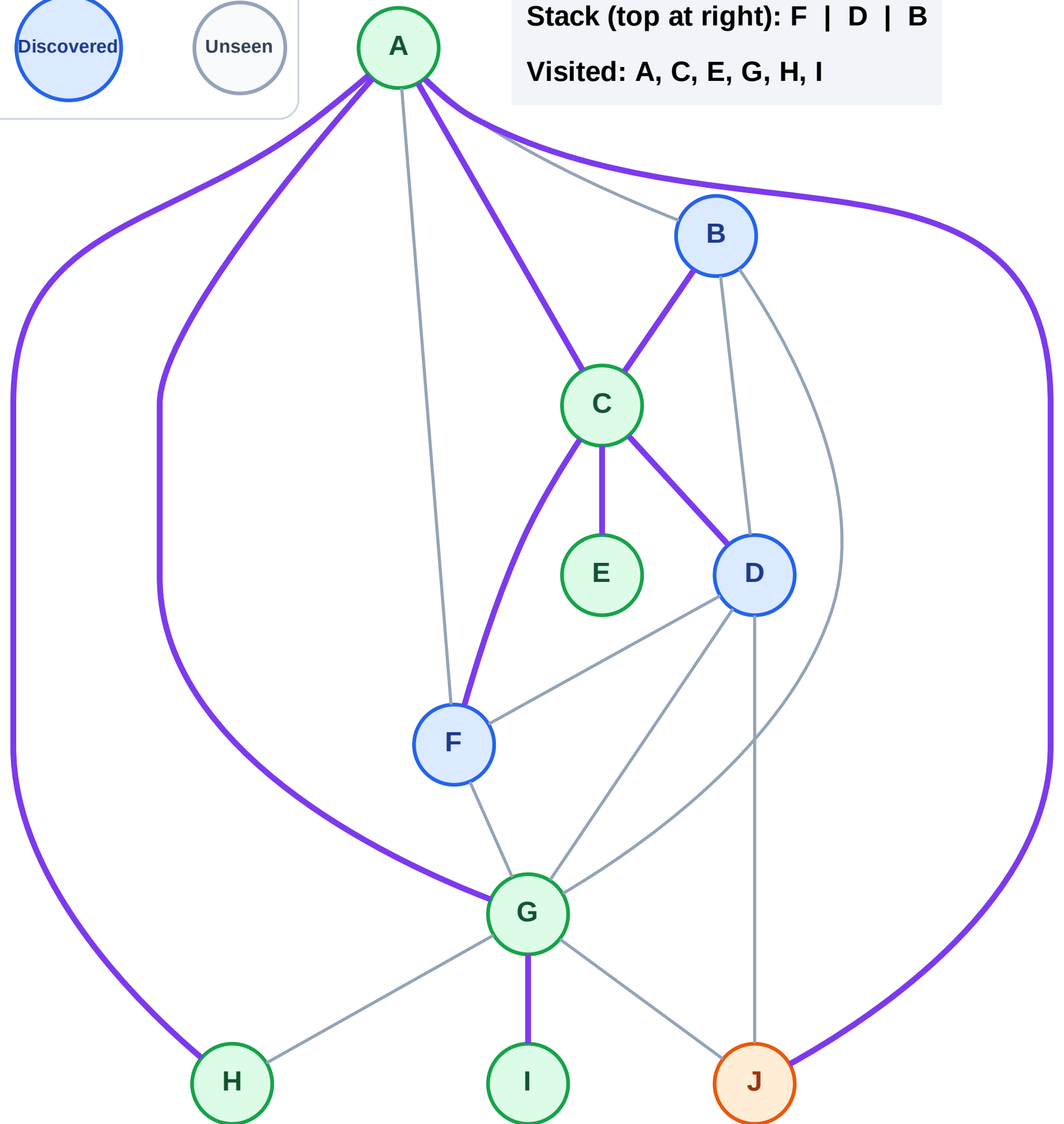
Discovered



Unseen

Stack (top at right): F | D | B

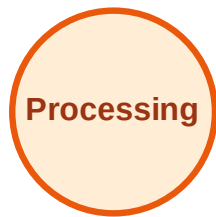
Visited: A, C, E, G, H, I



DFS Traversal

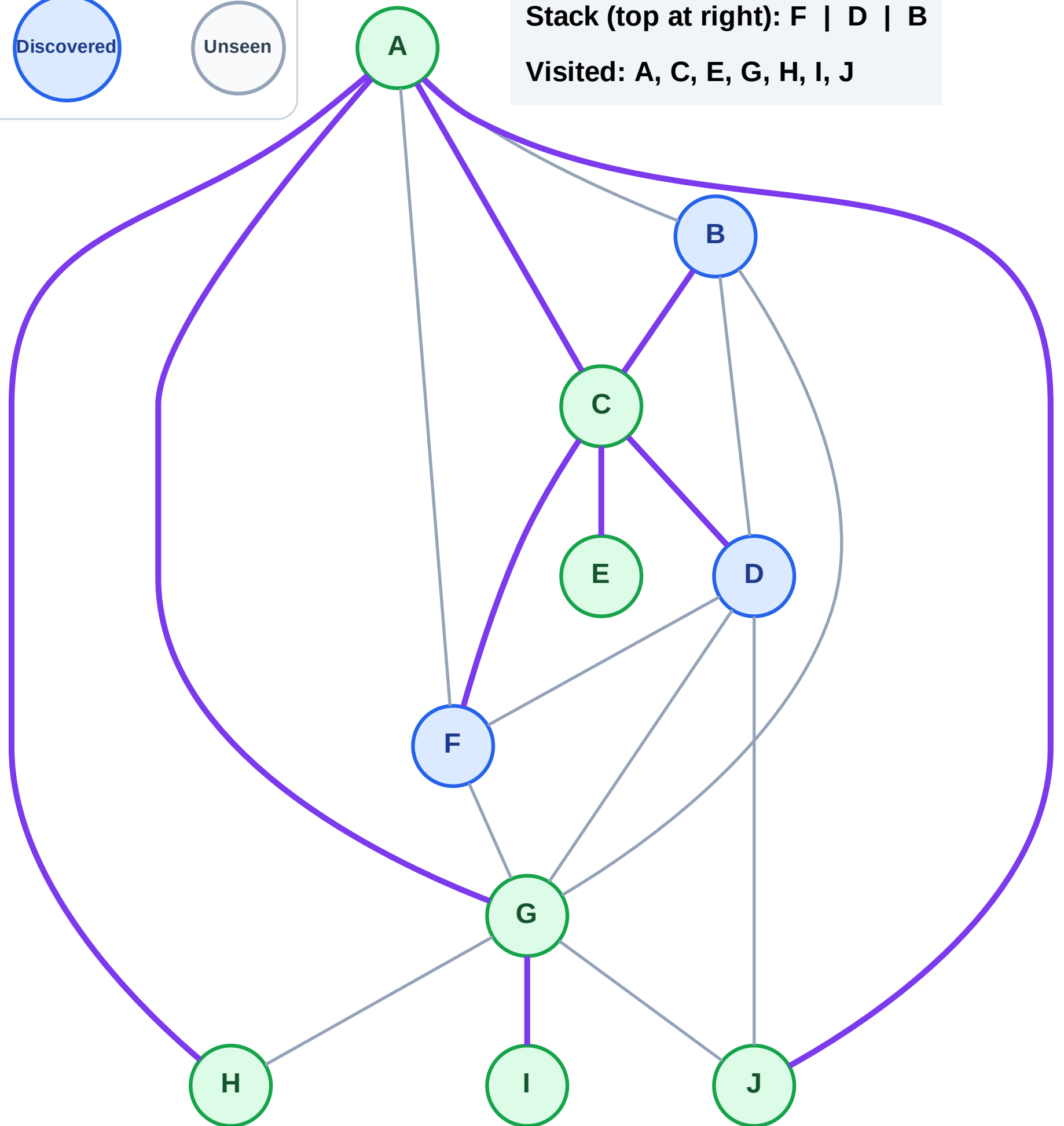
Step 15: No new neighbors from J

Legend



Stack (top at right): F | D | B

Visited: A, C, E, G, H, I, J



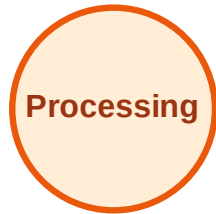
DFS Traversal

Step 16: Process node B

Legend



Complete



Processing



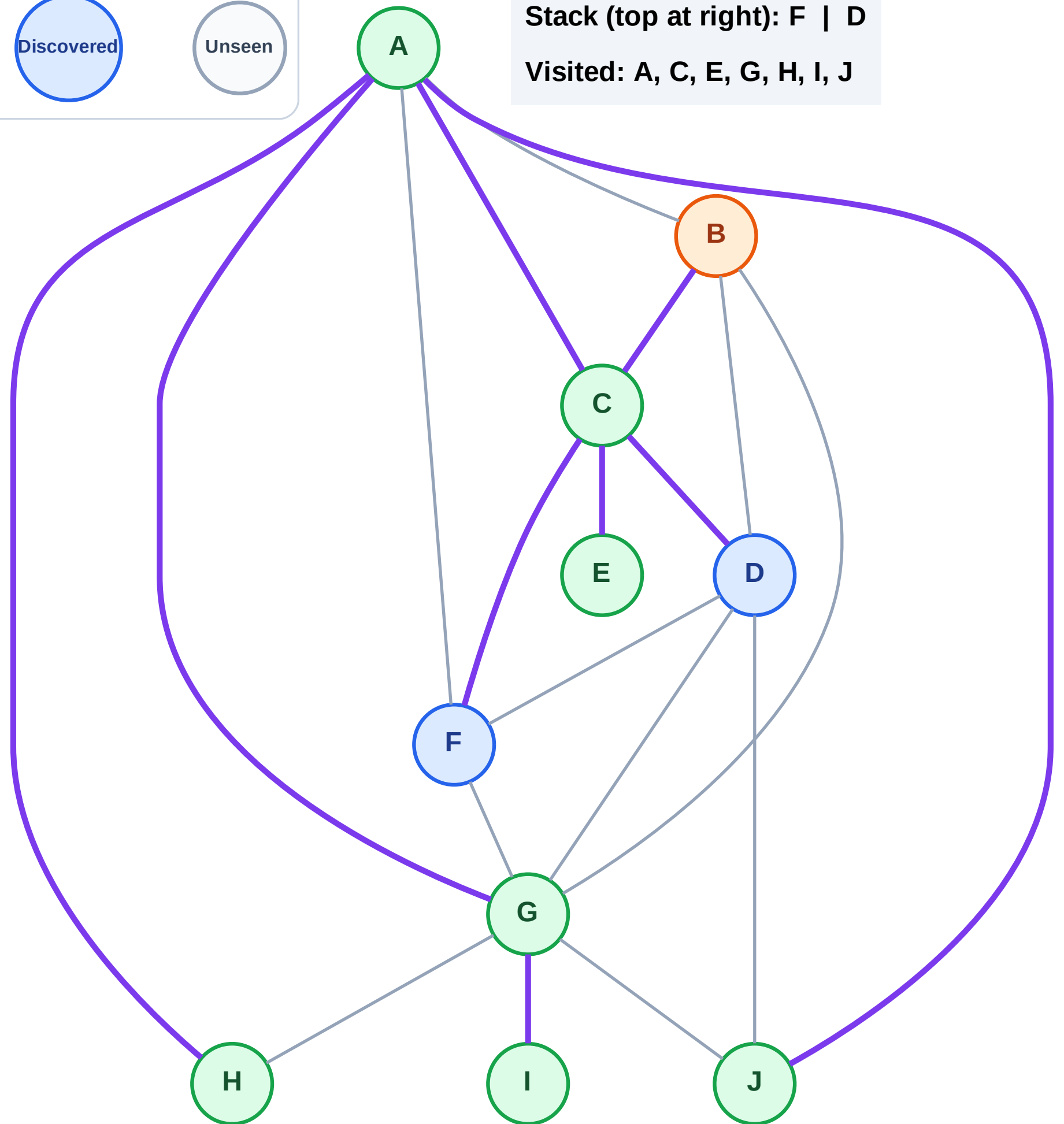
Discovered



Unseen

Stack (top at right): F | D

Visited: A, C, E, G, H, I, J



DFS Traversal

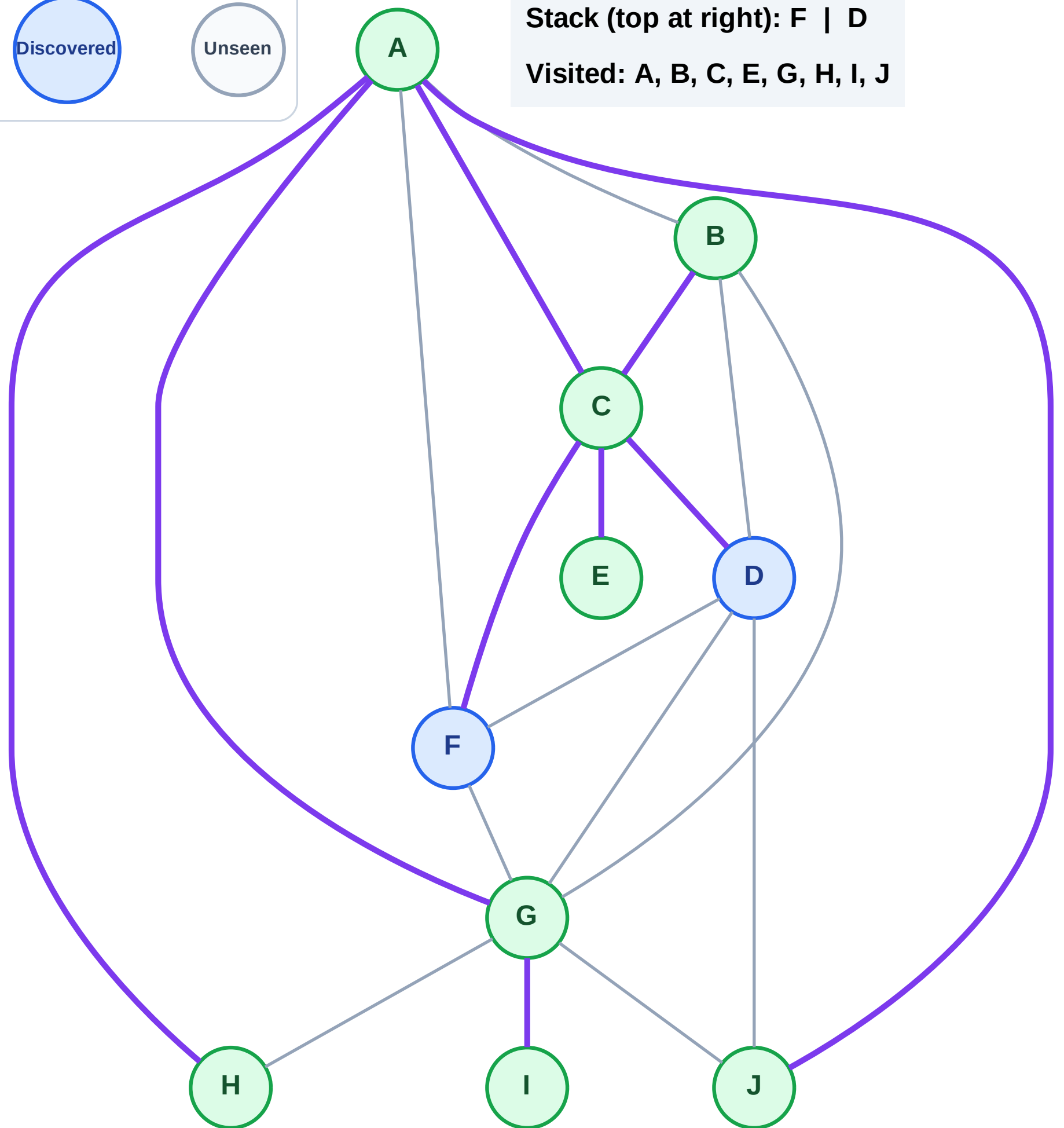
Step 17: No new neighbors from B

Legend



Stack (top at right): F | D

Visited: A, B, C, E, G, H, I, J



DFS Traversal

Step 18: Process node D

Legend

Complete

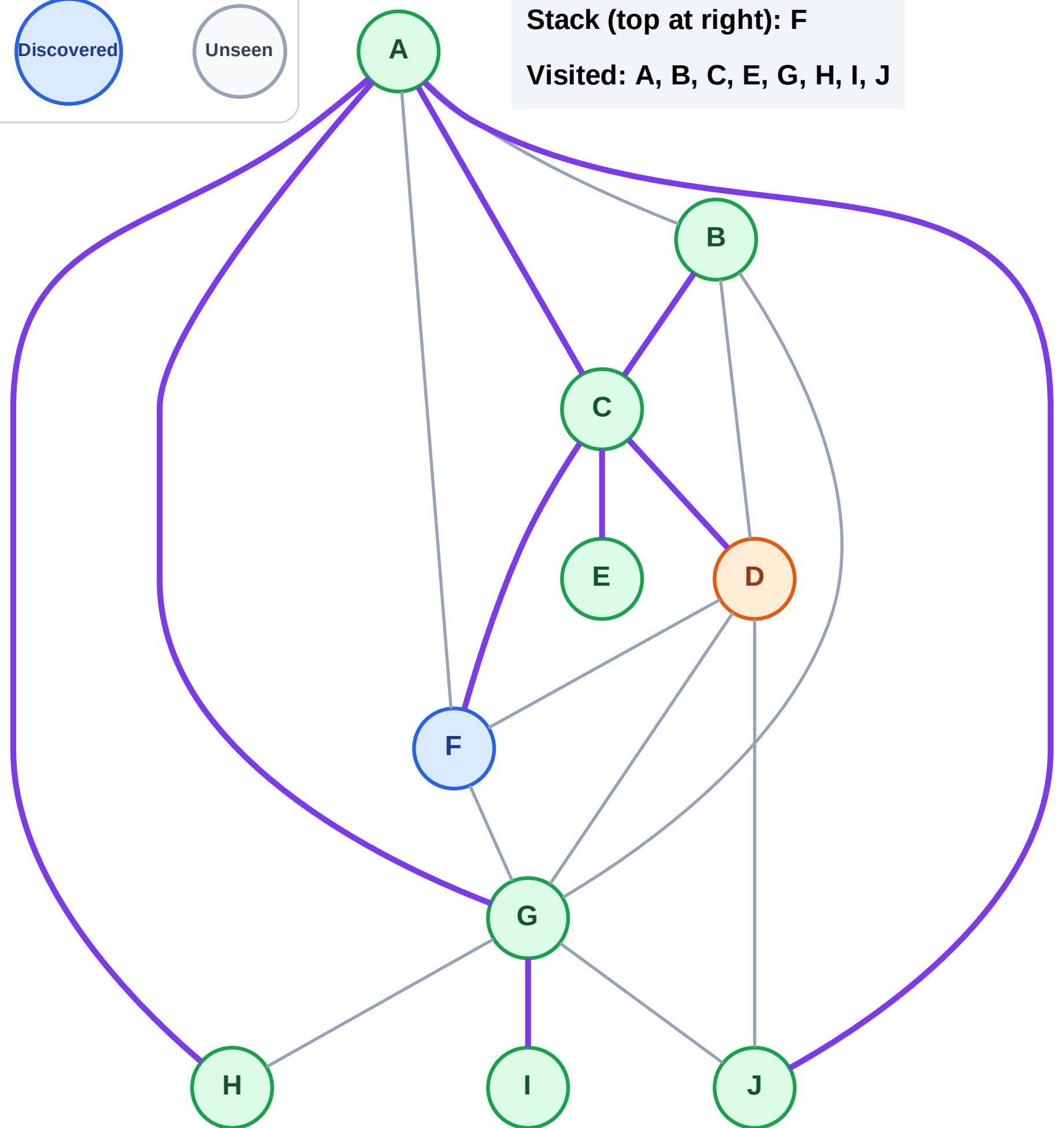
Processing

Discovered

Unseen

Stack (top at right): F

Visited: A, B, C, E, G, H, I, J



DFS Traversal

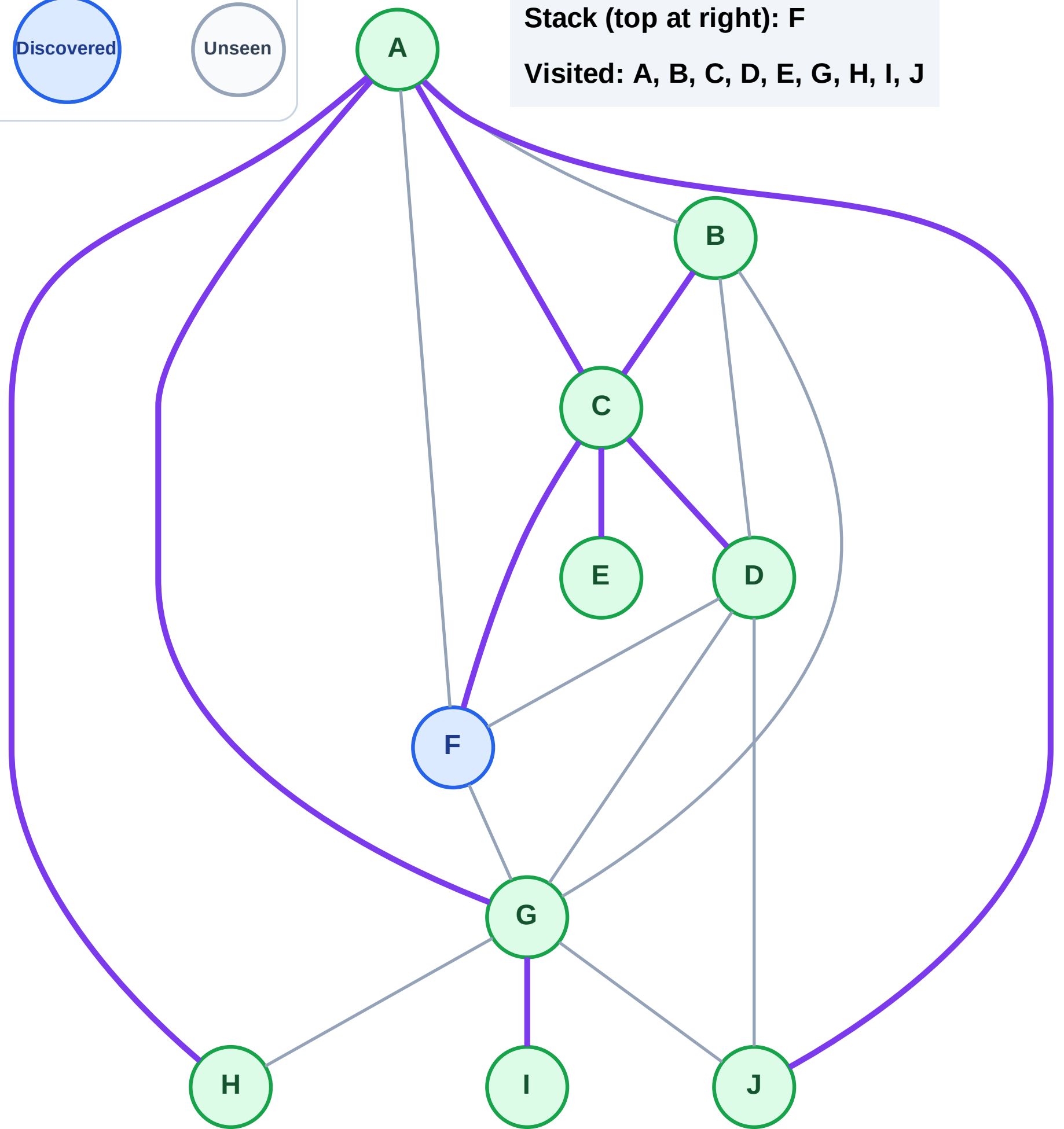
Step 19: No new neighbors from D

Legend



Stack (top at right): F

Visited: A, B, C, D, E, G, H, I, J



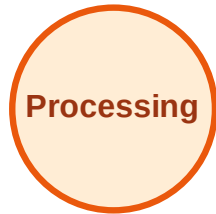
DFS Traversal

Step 20: Process node F

Legend



Complete



Processing



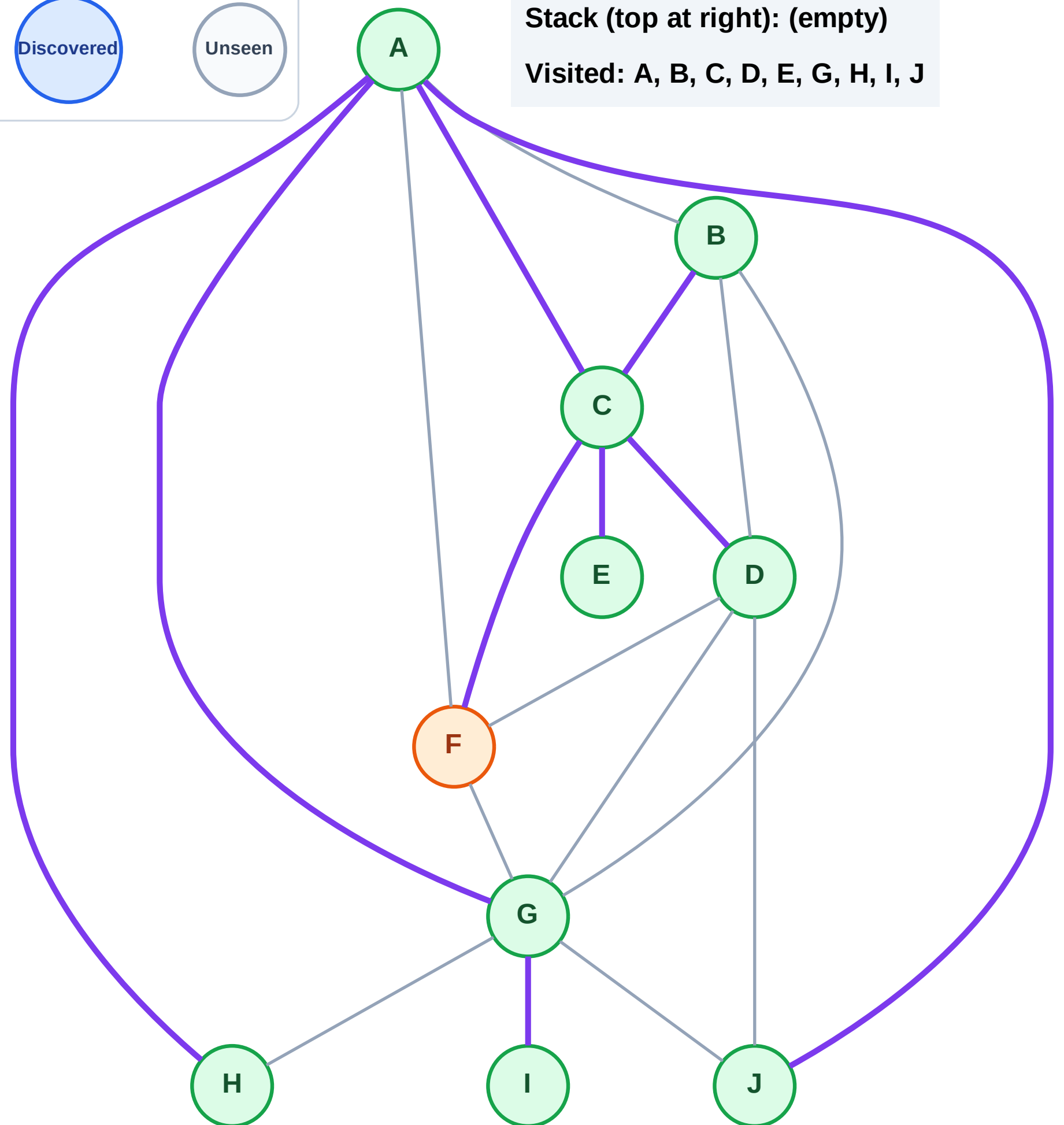
Discovered



Unseen

Stack (top at right): (empty)

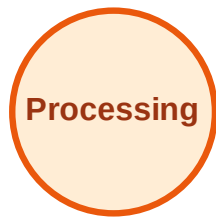
Visited: A, B, C, D, E, G, H, I, J



DFS Traversal

Step 21: No new neighbors from F

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

